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EDUCATION & JOB EXPERIENCE

EDUCATION:

BSC: BUTEX 36TH BATCH(W.P.E.)

MSC: BUTEX 5TH BATCH(ON GOING)(W.P.E.)

JOB EXPEIENCE:

COMPANY/UNIVERSITY	POSITION
1.EPYLLION KNITEX LTD (E.K.L)	EXECUTIVE IN DYEING PRODUCTION
2. NITER(D.U Tech Dept)	LECTURER

SYLLABUS

WPE 243: Wet Processing-1

Part-A

- 1.Water

  (Water and its importance in textile , source , concept of hardness , removal techniques, concepts of WTP in industry its importance and operations)
- 2.Detergent and Auxiliaries

  (Various auxiliaries their formation importance and mechanism on textile surface & uses area , properties and uses of various alkalis and acids in pretreatment, Detergents ,its type ,importance , use and mechanism)
- 3.Pretreatment

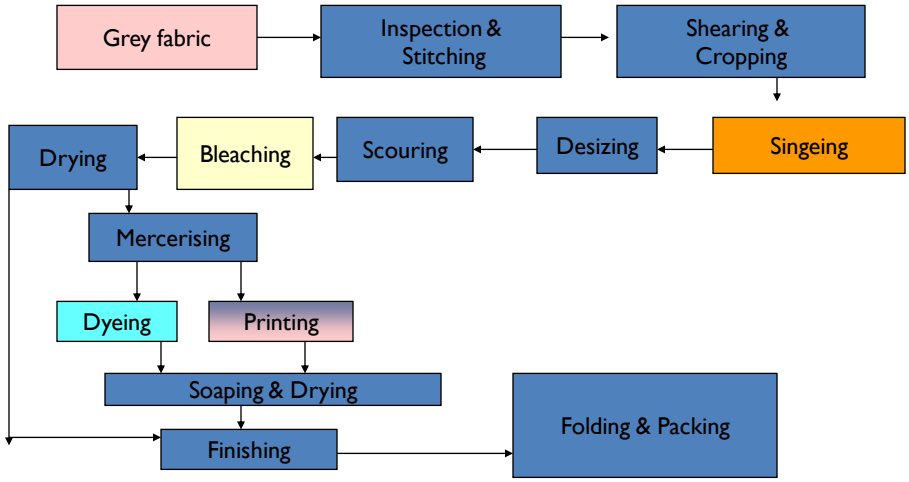
  (Chemistry of various impurities in fibre , concepts on singeing , desizing , scouring , bleaching, methods of applications in knit and oven fabric , pretreatment on cellulose and protein fibre , pretreatment machineries , modern concepts of pretreatment and industrial application estimation of pretreatment, damages on textiles due to pretreatment etc)

CHAP:1

INTRODUCTION



PROCESS SEQUENCES OF CHEMICAL PROCESSING

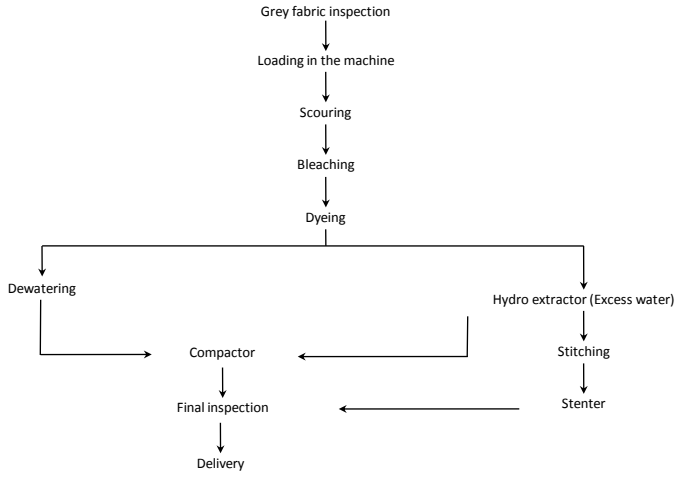


Preparation/Pretreatment

- Singeing
- Desizing
- Scouring
- Bleaching
- Mercerization

Typically a woven cotton fabric would be prepared by sequence of process as shown.
In case of knitting sizing step is not involved

Flow chart of wet processing for knit fabric:



Overview

- Inspection: Neps, warp end breakage, weft end breakage ,hole spot remove.
- Stitching: Smaller length of fabric becomes larger length of fabric.
- Singeing: Projecting or floating fibre remove.
- Desizing: Size material remove.
- Scouring: Using Alkali (NaOH, Na_2CO_3) for increasing absorbency of fabric.
- Bleaching: Permanent white.
- Dyeing: Color produce.Non localized process.
- Mercerizing: To increase the lusture of fabric.
- Printing: Specific color. Localized dyeing.
- Finishing: Soft finish for consumer satisfaction.
  Hard finish
  Wrinkle free finish

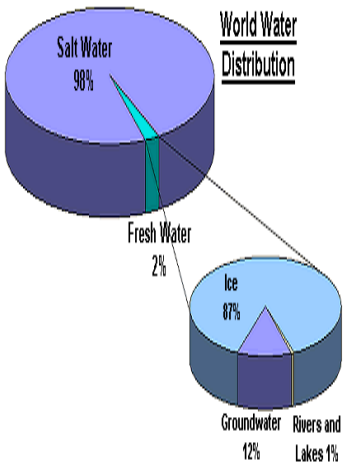
CHAP:2

WATER AND ITS TREATMENTS (W.T.P.)



WATER SOURCE

- **Water resources** are sources of water that are useful .Uses of water include agricultural , industrial, household, recreational and environmental activities. The majority of human uses require fresh water.
- 98 % of the water on the Earth is salt water and only three percent is fresh water; slightly over two thirds of this is frozen . The remaining unfrozen freshwater is found mainly as ground water. Water Is the life –line of textile industry



Source of water/type of water

- Rain water
- Surface water
- Subsoil water
- Deep well water

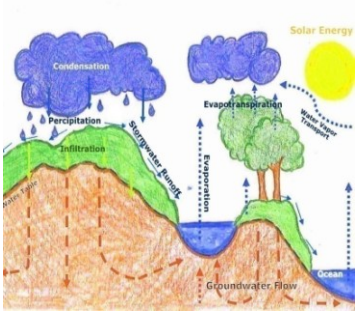
Rain water

- Rain, collected immediately after precitation, is the purest of all natural waters.
- It may contain traces of gases dissolved out of the atmosphere and possibly an almost infinitely small amount of finely divided solid matter derived from the air.
- It also contain dissolved or suspended impurities such as shoot traces of Sulphar di oxide or Sulphuric Acid ,CO_2, NH_3, NO_2 and other by products of industrialization.
- Suspended impurities present in it can be filtered by using sand bed.
- Suitable for boiling, washing and dyeing processes.



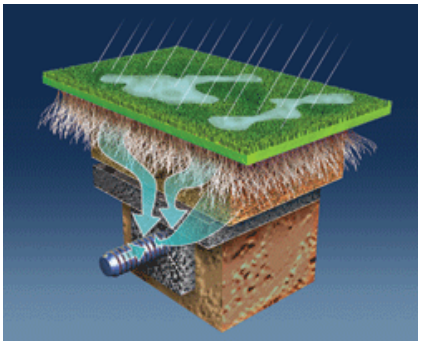
Surface water

- Surface water consists of rain water which has collected from streams, rivers or lakes.
- This type of water contains organic and inorganic matters which are dissolved in it & also contain suspended impurities.
- Then the Nutrifying bacteria will in time convert the organic substances into nitrates which are not objectionable in dyeing and finishing.
- Surface waters may receive considerable additions of dissolved mineral salts from shallow springs which feed the streams.
- It contains Chloride, Sulphate, Carbonate, Bicarbonate of Sodium, Pottasium, Calcium and Iron.
- Not suitable for dyeing & finishing.



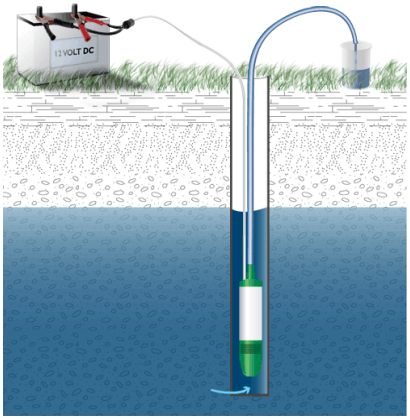
Subsoil water

- This type of water is collected from shallow springs and wells which are about 50 ft.(15m) or so deep.
- It is usually free from suspended impurities because it has been filtered by its passage through the soil. It will, however, contain dissolve organic matter.
- Subsoil water is often rich in dissolved carbon dioxide, a gas abundantly present in the skin of the soil.
- Subsoil waters are very variable with regard to the impurities which they contain.
- Not suitable for dyeing & finishing



Deep well water

- This type of water is obtained 500m below the surface. It is free from organic matters.
- The soluble impurities in water may be composed of a variety of substances. Soluble organic compounds, ammonium salts, nitrates and nitrites of animal or vegetable origin may be found. If they are present in considerable quantities, the sewage contamination is undesirable for many textile purposes.
- The presence of salts of calcium or magnesium in solution can be most undesirable in many finishing process.

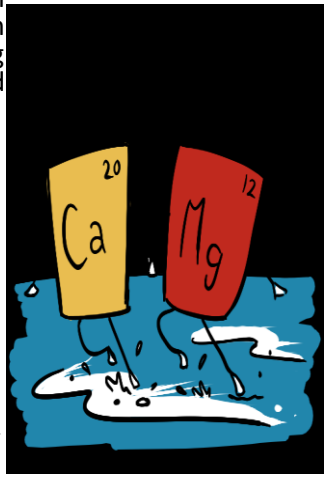


Hardness of water

- Generally soaps create foam in water, but in present of some materials the foam creation is reduced and need more soap for producing foam, and this condition of water is called water hardness.
- The presence of Calcium,Magnesium salt i.e. bicarbonates, sulphates, chloride in water is called causes of hardness of water. The water which contains these salts is called hard water. Hard water does not easily form lather with soap as the salt of Calcium and Magnesium react with soap to form insoluble organic salts.

$CaSO4 + 2RCOONa \rightarrow (RCOO)2Ca \downarrow + Na2SO4$

$MgSO4 + 2RCOONa \rightarrow (RCOO)2Mg \downarrow + Na2SO4$



Classification of hardness

1. Temporary hardness.
2. Permanent hardness.



Lime scale damage to Dishwasher interior



Lime scale build up on Heating Element



Lime scale has almost blocked these pipes



Blocked Shower head



Lime scale build up on tap



Lime scale removed from Hot Water Cylinders

Temporary hardness

- Temporary hardness is due to the presence of bi carbonates of calcium and magnesium. This type of hardness is called temporary hardness. Because it can be removed by easy means like boiling. When temporary hard water is boiled, the carbonates decompose with liberation of carbon-dioxide and precipitation of the insoluble Carbonates which are reformed.

$Ca(HCO_3)_2 \quad CaCO_3 \downarrow + CO_2 + H_2O$

$Mg(HCO_3)_2 \quad MgCO_3 \downarrow + CO_2 + H_2O$

- $MgCO_3$ is slightly soluble in water but heating will cause its hydrolysis into the much less soluble $Mg(OH)_2$.

- $MgCO_3 + H_2O \rightarrow Mg(OH)_2 + CO_2$

- So simple boiling and filtering of water remove temporary hardness.



Permanent hardness

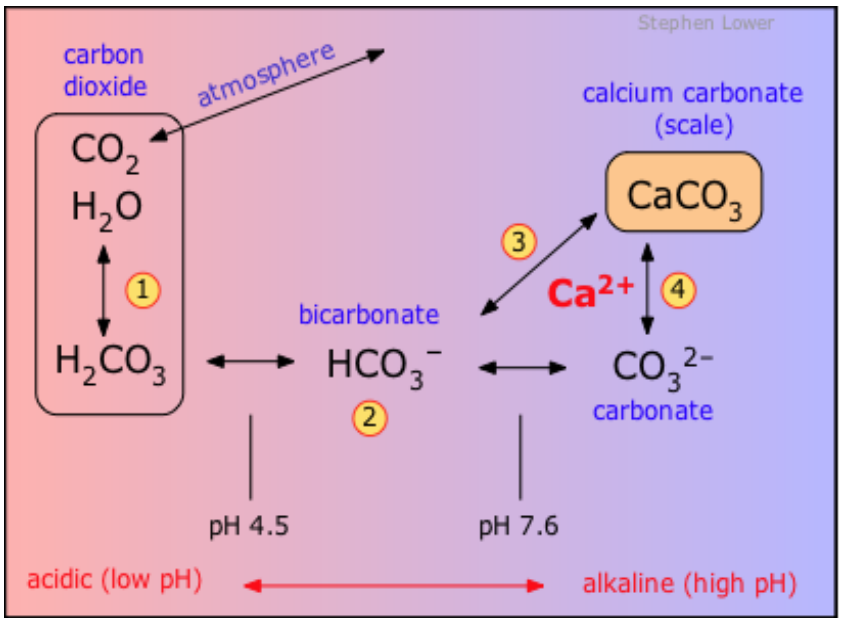
It is due to the presence of chlorides of Sulphates of Calcium and Magnesium. This type of hardness is called permanent hardness. These salts do not decompose on boiling. So permanent hardness can't be removed easily. It can be removed by lime when MgSO4 is responsible for hardness.

$CaSO_4 + Na_2CO_3 \rightarrow Na_2SO_4 + CaCO_3$

$MgSO_4 + Na_2CO_3 \rightarrow Na_2SO_4 + MgCO_3$

$MgSO_4 + Ca(OH)_2$ (Lime) $\rightarrow Mg(OH)_2 + CaSO_4$





UNITS OF HARDNESS

- Hardness is expressed by-
  1. PPM (Parts Per Million)
  2. In degrees (Grains/ gallon)

1. PPM: The number of grains of calcium carbonates which is present in one million grains of water is called PPM.1 grains of Calcium Carbonate present in 1 million grains water

2. In degrees: The number of grains of Calcium carbonates which is present in 70,000 grains of water.

Another unit of water hardness-

3. GPG – Grains Per U.S. Gallon
4. PP/ 100000
5. GPG imperial – Grains Per British Gallon

- Here,
  - » 1 U.S. gallon = 8.33 pounds
  - » 1 British gallon = 10 pounds (Used in our country)
  - » 1 grain = 1/7000
  - » pound;
  - » i.e. 7000 grains = 1 lb

Hardness Scales

1. German degree
2. French degree
3. American degree
4. British degree



Conversion factor of different water hardness scale

Scale	Hardness			
	USA	G	B	F
1º USA	1.0	0.056	0.07	0.1
1º G	17.9	1.0	1.25	1.79
1º B	14.3	0.8	1.0	1.43
1º F	10.0	0.56	0.7	1.0

Classification of water according to hardness

DESCRIPTION	TOTAL HARDNESS
Very soft	$0\text{-}4^0$
Soft	$5\text{-}8^0$
Mild	$9\text{-}14^0$
Fairy hard	$15\text{-}18^0$
Hard	$19\text{-}30^0$
Very hard	$>30^0$

From the above types of water, soft water with total hardness 5-80 is suitable for dyeing. In another cases like scouring we may use hard water.

Water hardness can also be noted as below:

Upto 50 PPM → Water is very soft
50 to 100 PPM → Water is moderately soft
100 to 150 PPM → Water is slightly hard
200 to 300 PPM → Water is hard
Above 300 PPM → Water is very hard

Definition of Different Hardness

1. **1º H (German) Hardness:** 10 mg CaO in 1 litre of water
2. **1º H (French) Hardness:** 10 mg CaCO3 in 1 litre of water
3. **1º H (English) Hardness:** 10 mg CaCO3 in 0.7 litre of water
4. **1º H (American) Hardness:** 1 mg CaCO3 in 1 litre of water

STANDARD/ QUALITY OF DYE HOUSE WATER

MINIMUM STANDARD	PERMISSIBLE CONCENTRATION
Color	Colorless
Smell	Odorless
PH value	Nature (PH 7.8)
Water hardness	Less than 50 dH
Dissolved solids	Less than 1 ml/L
Solids deposits	Less than 50 mg/ L
Organic substances	Less than 20 mg/ L
Inorganic salt	Less than 500 mg/ L
Iron (Fe)	Less than 0.1 mg/ L
Copper (Cu)	Less than 0.005 mg/ L
Nitrate (NO3)	Less than 50 mg/ L
Nitrite (NO2)	Less than 5 mg/ L

Iron and copper are responsible for the creation of spots on fabric. For those spots we can use 'spot removers'.

ESTIMATION OF TEMPORARY HARDNESS BY TITRATION BY TITRATION WITH HCL

For determining temporary hardness 200cc hard water is taken into a 500 cc bottle. Then few drops of methyl orange is added in it as an indicator. Now titration is carried out by adding 0.1N cold HCl until the yellow color of methyl orange turns colorless.

Here, each meal 0.1N HCl is equivalent to 0.005 gm of CaCO3. The associated reactions are as follows:

$$Ca(HCO3)2 + 2HCl \rightarrow CaCl2 + CO2 + H2O$$
$$Mg(HCO3)2 + 2HCl \rightarrow MgCl2 + CO2 + H2O$$

EXPRESSION:

Multiplying the required amount of 0.1N HCl for titration in cc by –

2.5 gives French hardness

1.78 gives English hardness

1.4 gives German hardness.

Permanent hardness can be found by deducting the temporary hardness from total hardness i.e.

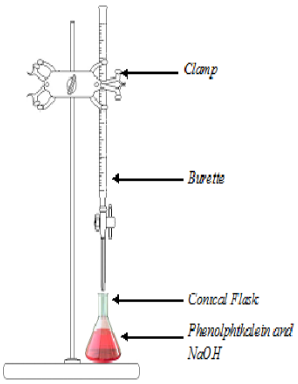


Fig: - Burette and clam with conical flask

- Permanent hardness = Total hardness – Temporary hardness

Estimation of total (permanent & temporary) hardness of supply water

Basic principle:

- Titration of sample water against standards (0.01M) EDTA solution

Preparation of 0.01M or 0.02N EDTA solution:

Molecular weight of disodium salt of EDTA
(CH2COOH)2 N2(CH2)2(CH2COONa)2.2H2O
= (12+1*2+12+16*2+1)×2 + 14*2+(12+2)*2+
(12+1*2+12+16*2+23)×2 + 2*18
= 118+ 28+28+162+36
= 372

Therefore,

In 1M solution of 1000ml contain 372 gm Na2-EDTA

In 0.01M solution of 1000ml contain 3.72 gm Na2-EDTA

In 0.01M solution of 100ml contain 0.372 gm Na2-EDTA

Process

- Preparation of ammonia buffer solution:

- 145ml of liquor ammonia (NH_4OH) of specific gravity 0.88+15gm NH_4Cl + distilled water to make 250ml solution to give a pH of 10.

Procedure:

- Add 1ml of buffer solution (NH_4OH+NH_4Cl) to 100ml of the original water sample. Add 3-4 drops of Eriochrome Black T indicator (0.2g dye in 15ml of triethanol amine + 5ml of ethanol)/ 1tablet (making powder) total hardness indicator.

- Titrate against 0.01M prepared EDTA solutions in burette until the color charges from wine red (or violet) to pure blue (or turquoise) with no reddish tone; then calculate the total hardness in terms of ppm of $CaCO_3$.

Calculation

TOTAL HARDNESS =

$$\frac{\text{Volume of 0.01M EDTA solution in ml}}{\text{Volume of sample water in ml}} \times 1000 \text{ ppm of } CaCO_3.$$

How does the water hardness affect the textile processing?

Desizing	Deactivate enzymes and makes it insolubilize some size materials like starch and PVA
Scouring	Combine with soap, precipitate metal-organic acids. Produce yellowing of off-white shades, reduce cleaning efficiency, and water absorption
Bleaching	Decompose bleach baths
Mercerizing	Form insoluble metal oxides, reduce absorbency and luster

How does the water hardness affect the textile processing?

Dyeing	Combine with dyes changing their shades, insoubilize dyes, cause tippy dyeing, reduce dye diffusion and hence results in poor washing and rubbing fastness.
Printing	Break emulsions, change thickener efficiency and viscosity, and those problems indicated for dyeing
Finishing	Interfere with catalysts, cause resins and other additives to become nonreactive, break emulsions and deactivate soaps

Problems associated with hard water

Scale formation in boiler

- Temporary hardness is rapidly converted to Calcium Carbonate and magnesium hydroxide in a boiler and in time accumulates in the form of scale on the inner surface of the shell or in the tubes.

$$Ca(HCO_3)_2 \rightarrow CaCO_3 + CO_2 + H_2O$$
$$Mg(HCO_3)_2 \rightarrow MgCO_3 + CO_2 + H_2O$$
$$MgCO_3 + H_2O \rightarrow Mg(OH)_2 + CO_2$$

Hence $\left[CaCO_3 + Mg(OH)_2\right]$ is called scale

- Heat loss by pipe scaling upto 40% for 20 mm scale

Problems associated with hard water

- **Reaction with soap:** The salts of Calcium and Magnasium (Permanent hard water) does not easily form lather with soap to form insoluble organic salts and becomes wastage of soap.

$$2RCOONa + MgSO_4 \rightarrow (RCOO)_2 Mg + NaSO_4$$
$$2RCOONa + CaSO_4 \rightarrow (RCOO)_2 Ca + Na_2SO_4$$

- **Reaction with dyestuff:** As dyestuff reaction hard water some dyestuffs are precipitated. So dyestuffs are spoiled. Hence shade which harm the quality off fabric.
- **Corrosion of boiler:** Corrosion can be a serious causes of wear in boilers if suitable water are not used. Dissolved oxygen in the presence of CO_2 is a common cause of corrosion especially affecting high pressure boilers. The CO_2 reacts with the iron, forming ferrous carbonate which in turn tends to hydrolyse to ferrous hydroxide.

$$Fe + H_2O + CO_2 \rightarrow FeCO_3 + H_2$$
$$FeCO_3 + H_2O \rightarrow Fe(OH)_2 + CO_2$$

Problems associated with hard water

Deposition on the goods during scouring

- Soap react with hard water and produce insoluble salts which deposit with the fabric during scouring. Insoluble salts do hard and unflexible the fabric which create problem in the next process.
- Textile industry is confronted with three main problems connected with water namely-
  - Problem of water of suitable quantity for processing of Textile products.
  - Supplies of water for boiler faced for power plant.
- Prevention of corrosion of Metal Tasks, Pipelines etc.

Water softening plant/ methods of water softening

1. Soda lime process.
2. Base exchange process (Permutit)
3. Demineralisation
4. Soda alum
5. Aeration
6. Chelation on sequestration.

Soda lime process

The main parts of the process are

1. Reagent tanks (Soda lime + Coagulants)
2. Reaction tank
3. Filter
4. Soft water storage tank.

- The soda lime & coagulants are entered in the reagent tanks. Predetermined amount of hard water is pumped into the reaction at the time of entering of reagents. The agitation is brought about by a large propeller. The agitation is increased to get more amount of ppm, steam is passed through the sideway pipe to increase the temp. of the mixer. When the precipitation is completed, the water is supplied to the filters to remove $CaCO_3$ & then finally to the soft water storage tank.
- The rate of precipitation may be increased by:
  - By increasing of temp. which hasten, crystallization & reduce stability.
  - By using an excess of reagent and stirring.
  - By bringing the water into contact with preformed particles of precipitate or grains of sand which can act as nucleus for the precipitation.
  - The lime soda [Na2CO3 + Ca(OH)2] and coagulant (NaAlO2) are metered into the reaction tanks together with a predetermined amount of hard water. Agitation is brought about in every tank by a large propeller. When sufficient time has elapsed for the precipitation to be completed the water passes through filters to the soft water storage.

THE RESULT

- By this process we can produce soft water with 50-100 ppm. But if temperature and agitation are increased water with 5-20 ppm hardness can be obtained

Soda lime process

In this process hydrated lime and sodium carbonate is used to remove the hardness.

For temporary hardness –

$$Ca(HCO3)2 + Ca(OH)2 \rightarrow 2\ CaCO3 + 2\ H2O$$

$$Mg(HCO3)2 + Ca(OH)2 \rightarrow MgCO3 + CaCO3 + 2\ H2O$$

$$MgCO3 + Ca(OH)2 \rightarrow Mg(OH)2 + CaCO3$$

For permanent hardness –

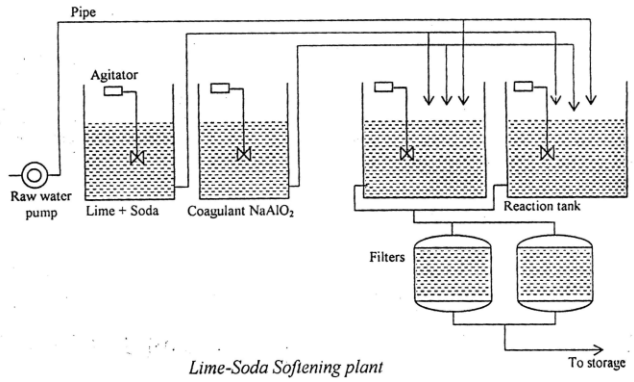
$$CaSO4 + Na2CO3 \rightarrow CaCO3 + Na2SO4$$

$$MgCl2 + Ca(OH)2 \rightarrow CaCl2 + Mg(OH)2$$

CaCl2 form is removed by –

$$CaCl2 + Na2CO3 \rightarrow 2\ NaCl + CaCO3$$

Soda lime process



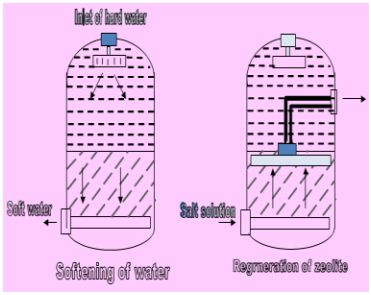
Lime-Soda Softening plant

Base Exchange process

- This method depends upon the use of Zeolite or Base Exchange complexes. The Zeolites are hydrated silicates of sodium and aluminum with a general formula. $(Na_2O)_X(Al_2O_3)_Y(SiO_2)_Z(H_2O)_n$.
- When Zeolites or base exchange complexes are brought in contact with hard water following reaction occur.
- For temporary hardness,
- $Ca(HCO_3)_2 + Na_2O.Z \longrightarrow CaO.Z + Na_2CO_3 + H_2O$
- $Mg(HCO_3)_2 + Na_2O.Z \longrightarrow MgO.Z + Na_2CO_3 + H_2O$
- For permanent hardness,
- $CaSO_4 + Na_2O.Z \rightarrow CaO.Z + NaSO_4$
- $MgSO_4 + Na_2O.Z \rightarrow MgO.Z + NaSO_4$
- In where Z is an abbreviation for the $Al_2O_3\ SiO_2\ H_2O$ part of Zeolite. The soft water obtained from this base exchange process is of $0 - 20^0$ hardness or levelness. After a long time the whole of the sodium in base exchange substance is replaced by calcium or magnesium, it is said to be exhausted because it will not soften any hard water more. Then it has to be generated.

Base Exchange process

- The zeolites are taken in the vessel as shown in figure with other required substances. When the hard water is passed through the inlet, comes in contact with zeolites, the water softened and soft water is collected from the downward outlet.
- When sufficient amount of hard water has passed then the supply of hard water is closed and then flow is reserved & beds of zeolites & other substances are cleansed. Then the cleansed is regenerated by passing 10% NaCl through the zeolites & the zeolites are regenerated again.
- Regeneration: $CaO.Z + 2NaCl \longrightarrow Na_2O.Z + CaCl_2$
- The $CaCl_2$ and residual NaCl are washed away and the regenerated $Na_2O.Z$ can be used to soften the hard water again.



Demineralization process

- The newer synthetic polymer ion exchangers are much more versatile than the zeolites and are widely used for water softening and demineralization. They are often called ion exchange resins. This reagent can remove all mineral salts to complete demineralisation of hard water. It has two types of ion exchanger – Cation exchanger and Anion exchanger

Hydrogen /cation exchanger

- In this process, sulphonated resin in hydrogen form is used.
- For water softening,

$CaCl_2 + H_2R \longrightarrow CaR + HCl$

(Resin of hydrated form)

$Ca(HCO_3)_2 + H_2R \longrightarrow CaR + +2\,H_2O + 2CO_2$

For regeneration, done by less then of 2%H_2SO_4

$CaR + H_2SO_4 \longrightarrow CaSO_4 + H_2R$(Regenerated hydrogen cation exchanger)

Demineralization process

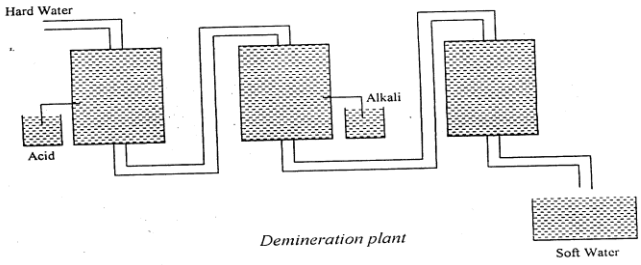
- **Anion exchanger**:
  Amino resin is used as an anion exchanger. & producing HCl is removed.

$HCl + HOR_1 \longrightarrow ClR_2 + H_2O$(Soft water)
(Resin of Hydroxide form) Precipitation

Regeneration reaction, 1% Solution of Caustic soda
$ClR_2 + NaOH \longrightarrow Na_2SO_4 + HOR_2$(Regenerated resin)
The regeneration is performed by using alkali

Demineralization process





Demineration plant

CHAP:3

Soap , Detergent and Surfactants



Soap

- Soaps are the sodium and potassium salts of the long chain carboxylic acid.A soap molecule consists of a long hydrocarbon chain (composed of carbons and hydrogens) with a carboxylic acid on one end which is ionic bonded to metal ion usually a sodium or potassium.
- A soap has a large non-ionic hydrocarbon group and an ionic group COO-Na

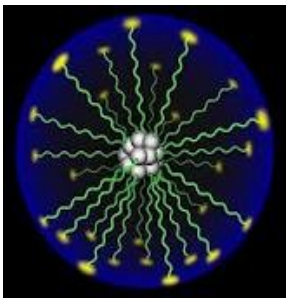
- **EXAMPLES OF SOAPS**

- **Sodium stearate** (Chemical formula: $C_{17}H_{35}COO^-Na^+$)
- **Sodium palmitate** (Chemical formula: $C_{15}H_{31}COO^-Na^+$)
- **Sodium oleate** (Chemical formula: $C_{17}H_{33}COO^-Na^+$)

Soap manufacturing process

- The process of making soap by the hydrolysis of fats and oils with alkalies is called saponification.
- Soap is made by heating animal fats or vegetable oil with concentrated sodium hydroxide (NAOH).
- Fat or Oil + NaOH → Soap + Glycerol

MICELLES – SOAP MOLECULES

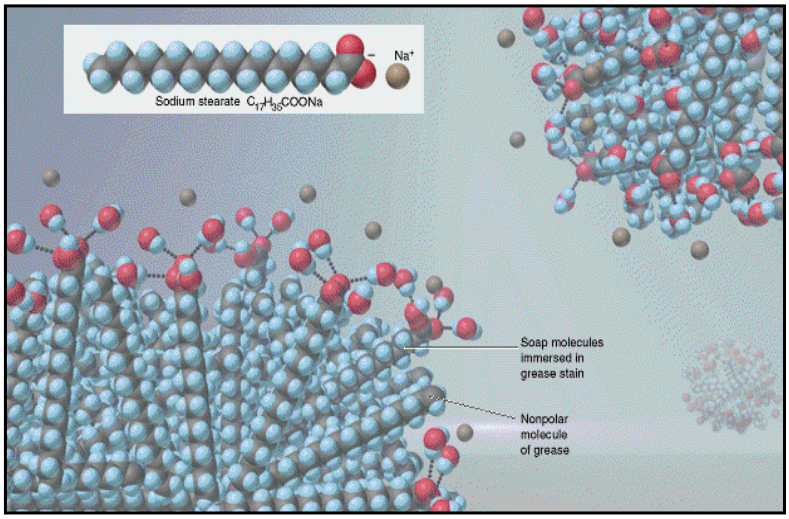


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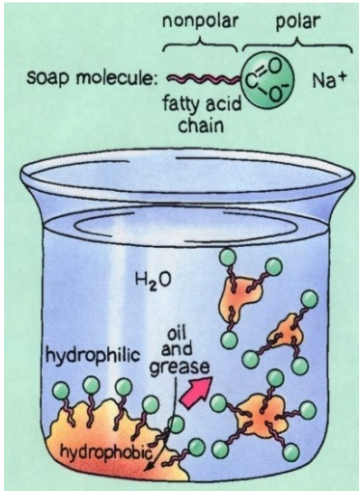
- A soap molecule has two ends with different properties-

1. A long hydrocarbon part which is hydrophobic (i.e. it dissolves in hydrocarbon).
2. A short ionic part containing COO-Na+ which is hydrophilic (i.e. it dissolves in water).

WORKING OF MICELLES



MECHANISM OF SOAPS

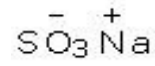


- When a dirty cloth is put is put in water containing soap than the hydrocarbon ends of the soap molecule in the micelle attach to the oil or grease particles present on the surface of dirty cloth. In this way the soap micelles entraps the oily particles by using the hydrocarbon ends. The ionic ends of the soap molecules remain attached to the water when the dirty cloth is agitated in soap solution. The oily particles presents on its surface gets dispersed in the water due to which the cloth gets clean.

Selection of soap

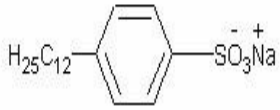
- Soap should contain at least 62% of fatty acid and between 6.5% and 8.5% combined alkali expressed as Na_2O.
- No free fat, unsaponifiable oil or free caustic should present, free caustic alkali denotes careless in manufacture but sodium carbonate or silicate are often added purposely to make the soap harder or to improve its detergent action.
- Salts such as Sodium Chloride and Sodium Sulphate should only be present in small quantity.
- It should show a high degree of surface affinity.
- The most important factors which determine the properties of soaps in the number of carbon atoms in the aliphatic chain. If there are less than 9(CH_2) group, the soap will have virtually no detergent power

DETERGENTS

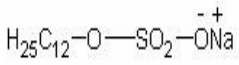
- Detergents are the sodium salts of long chain benzene sulphuric acids.
- Detergents are primarily surfactants, which could be produced easily from petrochemicals. Surfactants lower the surface tension of water, essentially making it 'wetter' so that it is less likely to stick to itself and more likely to interact with oil and grease.
- The ionic group is in a detergent is 

EXAMPLES OF DETERGENTS

- Two basic examples of well-known detergents of the sulphonate group or the sulphate group are:

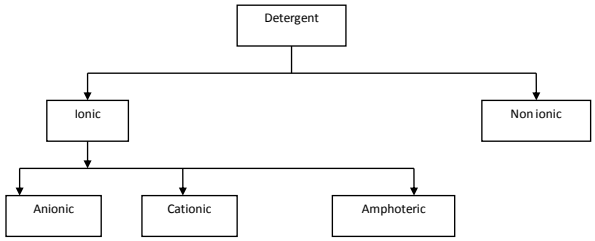


Sodium p-dodecyl benzene sulphonate



Sodium lauryl sulphate or
Sodium n-dodecyl sulphate

Classification



Anionic detergent

When the detergents are ionised into anions and cations but the anion is the dominating ion in the solution. Therefore the surfactant is called an Anionic surfactant. e.g. Soap.

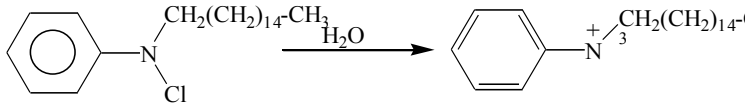
When Sodium Stearate is dissoved in water, it ionises as follows:

$$CH_3(CH_2)_{16}COONa \longrightarrow CH_3(CH_2)_{16}-COO^- + Na^+$$

Among the ions, anions $CH_3(CH_2)_{16}COO^-$ is comparatively very large to Sodium ion. So anion acts as dominating ion here. This Sodium stearate is called Anionic stearate

Cationic detergent

Ionic surface active agents which produce cation as dominating ion when dissolved in water is called Cationic detergent. e.g. Catyl pyridinium chloride. When C atyl pyridinium chloride is dissolved in water it consists as follows:

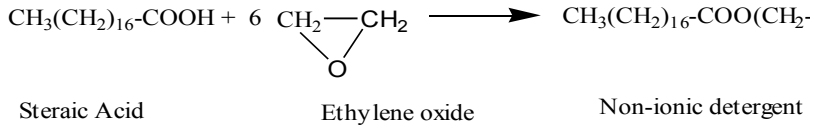


Among two ions cation is very large compound to the chloride ion. There fore cation behaves as a dominating ion in case of cat ionic surface active agents.

Non ionic detergent

Surface active agent which are soluble in water and get oriented at the surface of the solution and reduce surface tension of water .Non ionic detergents do not generally ionic when dissolve in water hence they are called "Non -ionic detergent"

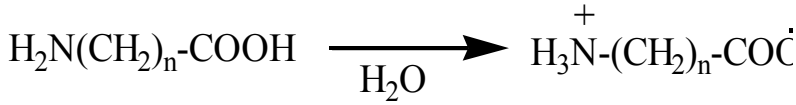
For example; when one mole of Stearic acid is condensed with six moles of Ethylene oxide a polyethe r is formed:



Steraic Acid | Ethylene oxide | Non-ionic detergent

Amphoteric detergent

Detergents when dissolved in water, ionise and produce large segments carrying both anionic and cationic ions(These segments are called zwitter ions). Thus amino carboxylic acids in which amino and carboxylic groups are present at the molecular chain ends dissolved in water to give zwitter ions.

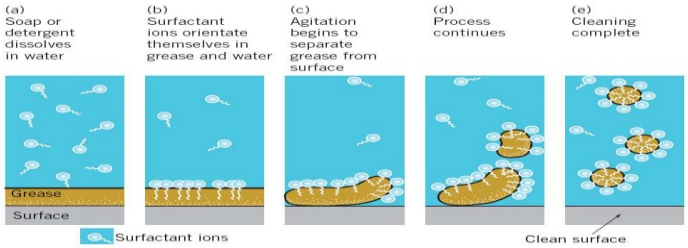


Special characteristics of Amphoteric detergents

- In alkaline solution: P^H>7
- They behave like Anionic detergent.
- When Acidic solution: P^H<7
- They behave like Cationic detergent
- When Neutral solution: PH=7
- They behave like Non ionic detergent
- Amphoteric surface active agent possess affinity towards wool and cellulose fibres
- They have lubricating properties

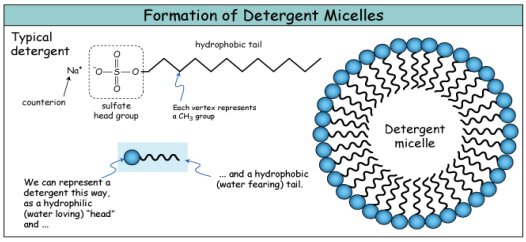
Mechanism of detergent

- A surfactant molecules is divided into two ion when dissolved in water-
- Hydrophillic head
- Hydrophobic tail.
- **Hydrophobic (tail)** are water avoiding and try to get away of the water. i.e. they protruding away from the water surface and hydrophilic(head) try to dissolve in water and just dipping at the water surface. As a result, the hydrophobic (tail) reduced the surface tension and the surface characteristics altered.
- As for example:
- Sodium stearate ($C_{17}H_{35}COONa$) dissolved in water and divided by following two segment-
- The aliphatic chain ($C_{17}H_{35}$) are hydrophobic (tail) and repelled by the water.
- And the electronegative (COO^-) carbonilic ion are hydrophilic (head) and just dipping at the water surface. As a result, the hydrocarbon chains (tail) reduces the surface tension and creates a force opposite to the inward pull on the water molecules. Then the most of ion get oriented on the surface between the water and air(/oil).



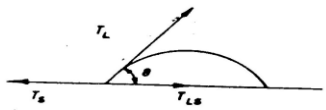
Mechanism of detergent

- When the fatty acid components of the (soap/detergents) compounds is of low molecular weight the hydrophilic (head) can pull the hydrophobic(tail) into the water, but the high number of carbon atoms prevent this force oppositely.
- A lowest number of carbon atoms necessary to show/manifest surface activity is of (C_9-C_{19}). The lowest series, which can be classified as good detergents are the [$CH_3(CH_2)_{10}$.COONa]
- When a fibre covered with a layer of oil or wax, is immersed in a warm aqueous solution of soap and it will be seen that the film bounches up and ultimately collects itself into spherical globules which are very easily detached.



Mechanism of detergent

- In a drop of oil, lenticular in foam, adheres to a fibre, its actual shape will be determined by the inter action of the following three forces-
- T_L= The surface tension between the drop and the surround in aqueous phase.
- T_{LS}= The surface tension between the fibre and the oil.
- Ts= The surface tension between the fibre and the water plane any solute which it may contain.
- In the figure, Any increase in (T_{LS}+T_L) in relation to Ts will decrease the area of contact between the oil and fibre. The result of this is that the shape of the oil drop will approach more and more to that of a sphere.
- is the contact angle which is drawn at the contact point of solid surface and oil drop. When the forces are equilibrium their relationship-
- $T_S = T_{LS} + TLCOS$
- $COS = (T_S - T_{LS})/T_L$
- $T_L = T_S - T_{LS}$ When $= 0^0$
- To fulfil this condition, the solid surface must be covered with that phase of the lowest surface tension. If Ts is greater than T_{LS},COS will be positive and will be less than 90^0. If is greater than 90^0 the value of COS will be negative and then T_{LS} will be greater than T_S. It indicates that the aqueous phase e will displace the oil.





DIFFERENCES BETWEEN SOAPS AND DETERGENTS

❑ SOAPS

- They are metal salts of long chain higher fatty acids.
- These are prepared from vegetable oils and animal fats.
- They cannot be used effectively in hard water as they produce scum i.e., insoluble precipitates of Ca2+, Mg2+, Fe2+ etc.

❑ DETERGENTS

- These are sodium salts of long chain hydrocarbons like alkyl sulphates or alkyl benzene sulphonates.
- They are prepared from hydrocarbons of petroleum or coal.
- These do not produce insoluble precipitates in hard water. They are effective in soft, hard or salt water.

Chap:4
SINGEING



Fig. : Typical Singing Machine

SINGEING

The verb 'singe' literally means 'to burn superficially'. Technically, singeing refers to the burning-off of. Loose fibres not firmly bound into the yarn and/or fabric structure. Singeing is an important part of pretreatment. This is the burning off of protruding fiber ends from the surface of the fabric. If not done properly, unclear print patterns, mottled fabric surfaces, and pilling results.

- Loose yarns not firmly bound into the fabric structure;
- Protruding fibre ends sticking out of the textile yarns and/or fabrics.

Textiles materials are most commonly singed in woven or knitted fabric form or in yarn form.



Singeing Definition

If a fabric is to have a smooth finish, singeing is essential. Singeing is a dry process used on woven goods that removes fibers protruding from yarns or fabrics. These are burned off by passing the fibers over a flame or heated copper plates. Singeing improves the surface appearance of woven goods and reduces pilling. It is especially useful for fabrics that are to be printed or where a smooth finish is desired. Pollutant outputs associated with singeing include relatively small amounts of exhaust gases from the burners.

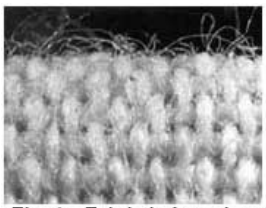


Fig. 1a. Fabric before singeing

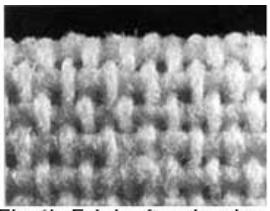


Fig. 1b. Fabric after singeing

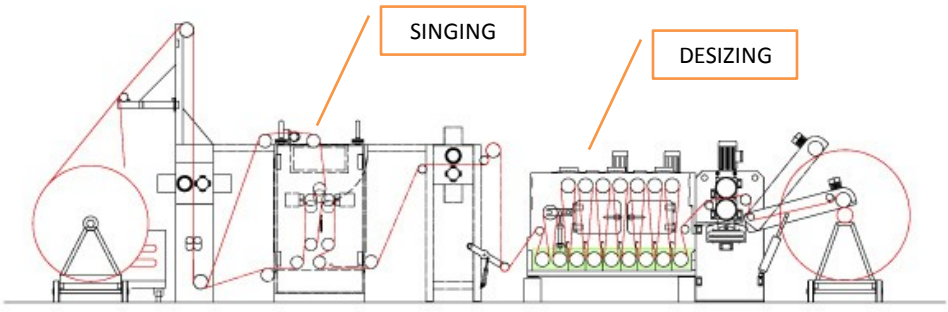
(Pic. by Osthoff-Senge)

SINGEING MACHINE



Fig. : Typical Singing Machine

SINGING DESIZING



SINGEING OBJECTIVES & ADVANTAGES

- Singeing of a fabric is done in order to obtain a clean fabric surface which allows the structure of the fabric to be clearly seen.
- Fabrics, which have been signed, soil less easily than un-singed fabrics.
- The risk of pilling, especially with synthetics and their blends, is reduced in case of singed fabrics.
- Singed fabrics allow printing of fine intricate patterns with high clarity and detail.
- The risk of skitter dyeing with singed articles dyed in dark shades is considerably reduced, as randomly protruding fibers are removed in
- Singeing which could cause diffused reflection of light.

SINGEING OBJECTIVES & ADVANTAGES

- Cotton materials are valued for their smooth appearance. After the formation of fabric it has a fuzzy or hairy appearance due to projecting fibers, thus affecting the luster and smoothness cotton is known for.
- Unsigned fabrics are soiled easily
- The protruding fibers obstruct the subsequent dyeing and printing process
- Goods which are to be mercerized are signed to maximize the luster
- In fabrics of polyester and cellulosic fiber blends singeing is the best method to control pilling, sometimes double singeing is done to minimize the pilling.

PRECAUTION DURING SINGEING

1. The fabric to be singed should be dry as wet fabric tend to scorch more readily than dry.
2. Uneven singeing may cause streaks on fabric or bubbles when the fabric is finished.
3. Improper singeing may lead to loss of 75 % loss in tensile strength loss in warp direction.
4. The fabric should not contain any acid releasing salt,which may release acid on heating and tender the fabric.
5. Stopping the machines may cause bars on the fabrics.
6. Singeing may cause hardening of the size thus leading to difficulty in its removal.
7. Possibility of thermal damage to temperature sensitive fabrics.
8. The burning characteristics of fibers must be taken into account when this process is applied, as heat-sensitive fibers melt, forming tiny balls on the surface of the fabric. These balls interfere with dye absorption, so that, as a general rule, heat sensitive fibers would be singed after dyeing or printing

TESTING SINGEING EFFECTIVENESS

The effectiveness of singeing process can be checked by one or more of the following:

- By looking at the singed fabric with magnifying glass and comparing its hairiness with that of the un-singed fabric. A well-singed fabric shows less hairiness.
- By testing the singed fabric for pilling performance and comparing it with that of the un-singed fabric. A well-singed fabric gives less pilling.
- By sticking and removing a sticking tape on the singed fabric and observing the number of fibres attached to the sticking side of the tape. A well-singed fabric results in less number of fibers sticking on the tape.
- Noticing the feel or handle of the singed fabric. An over-singed fabric may give a harsher feeling.

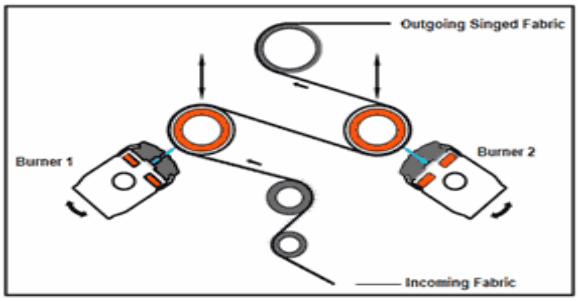
Fabric Singeing

There are three main types of singeing machines:

1. Gas singeing machine,
2. Plate singeing machine,
3. Rotary-cylinder or Roller singeing machine

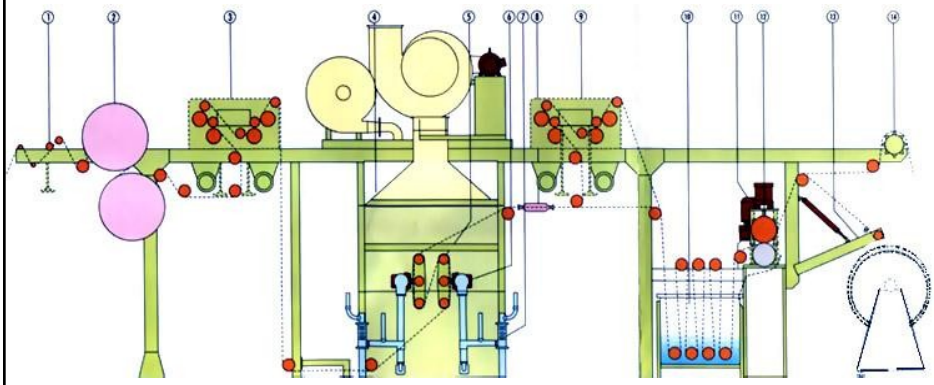
GAS SINGEING MACHINE

In this type of singeing machine, the fabric passes over a burning gas flame at such a speed that only the protruding fibres burn and the main body of the fabric is not damaged by the flame. This is the most common type of machine used for singeing fabrics as well yarns





Line Diagram of Gas Singeing Machine <u>for Woven Fabrics</u>



Main parts of a sigeing machine

i. Tension unit with guide rollers & bars
ii. Pre-drying cylinder (optional)
iii. Pre-bushing unit with suction
iv. Singeing chamber with exhaust
v. Water-cooled roller
vi. Water-cooled burner
vii. Carburettor with blower
viii. Steam quenching device
ix. Post-brushing with suction (optional)
x. De-sizing unit
xi. Squeezing mangle
xii. Main drive (A.C. motor with frequency controller)

Most singeing machines consist of: an entry section; a pre-brushing/cleaning section; a burner section; a post-brushing/cleaning section; an impregnator; and an exit/batching section



Fig. 5a. Brushing Roller



Fig. 5b. Beating Roller

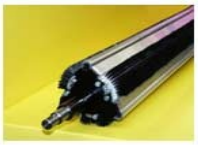


Fig. 5c. Combined Brushing & Beating Roller

GAS SINGEING MACHINE

- A gas-singeing machine is normally employed. The gas-singeing machine is typically equipped with four burners, and is capable of singeing one or both surfaces of the fabrics.
- A water-cooled roller is provided at a location opposite to the burners, thereby enabling singeing to be performed without undermining the strength of even thin fabrics. It is important to set a drain temperature of the water-cooled roller in a range of 50°C to 55° C.
- Cautions are required because a dew-point is generated when the water-cooled roller is cooled down too much, and results in increased amount of remaining fuzz or adhered carbon.
- The fabric feed speed is preferably set at around 100 to 150m/minute.

ADVANTAGE/ DISADVANTAGE OF GAS SINGEING

ADVANTAGE

- Both sides singeing in this process.
- It is a standard process and ideal singeing.
- It is a continuous process.
- Fabric becomes very suitable for subsequent process

DISADVANTAGE

- Not suitable for synthetic fibre.
- Dirty burner can produce spot on the fabric.
- Due to inconsistent speed, fabric may get burnt

IMPORTANT GAS SINGEING PARAMETERS

1.Flame Intensity

Together with the supply and control units for gas-air mixture, burners comprise the most important part of any singeing machine. The flame intensity of the singeing burners is based on the amount and the outlet speed of the gas-air mixture leaving the burner slots. Besides having high thermal energy, flame also has considerable mechanical energy. All the thermal and mechanical energy of the flame is directed onto the fabric during singeing. The temperature of the flame at the mouth of the burner is in the range of **1250 to 1300C**. The speed of the flame at the burner outlet may be between **15 and 35 meter** per second. The flame intensity usually lies between **5 and 20 bars**.

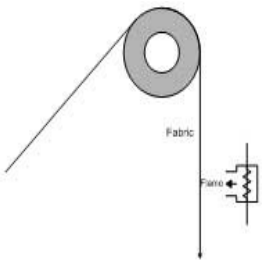
2. Fabric Speed

The fabric speed in the singeing machine is usually in the range of **50-160 m/min** depending on fabric (gram per square meter) weight and fibre blend. For heavier fabrics, the speed is kept slower as compared to lighter weight fabrics

3. SINGEING POSITION

A) Singeing onto free-guided fabric

This is the most intensive singeing position with highest efficiency. In this position, the flame bounces onto the free-guided fabric at right angles. This position is usually recommended for singeing of fabrics with all natural fibres (e.g. cotton), regenerated fibres and blended fabrics, which have been tightly woven and have weights over 125 g/m2.



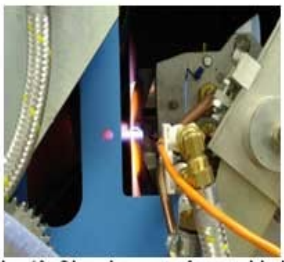
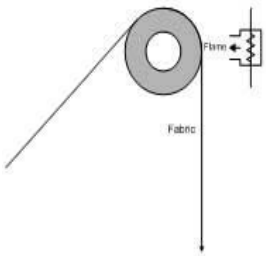


Fig. 10. Singeing onto free-guided fabric

B) Singeing onto water-cooled roller

In this position, the flame bounces at right angles onto the fabric while the fabric passes onto water-cooled guide roller. This position is usually recommended for all blended and synthetic fabrics as well as for fabrics having weights less than **125 g/m2** and fabrics with open structure.



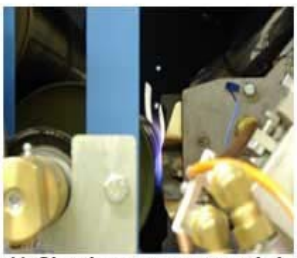


Fig. 11. Singeing onto water-cooled roller

C) Tangential Singeing

In this position, the singeing flame falls on the fabric tangentially. This position is usually recommended for very light weight and sensitive fabrics as well as fabrics with broken filaments.

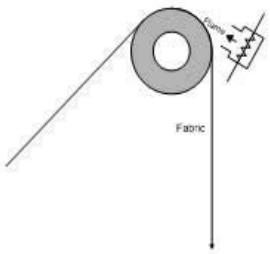




Fig. 12. Tangential singeing

4. Distance between Flame Burner and Fabric

The distance between the burner and the fabric is usually in the range of **6-8mm** but it can be adjusted in a range from **6-20mm**.

5. Flame Width

All good singeing machines come with a provision of flame width adjustment according to the width of the fabric. This is essential to optimize the gas economy.

ESSENTIAL CONDITIONS FOR GOOD GAS SINGEING

1. A flame with high mechanical & thermal energy to quickly burn thermoplastic protruding fibres (e.g. polyester) without any molten beads formation.
2. A homogeneous flame with uniform mechanical & thermal energy to result in uniform singeing
3. An optimal flame/fabric contact time to neither results in incomplete not over-singeing.

Common problems in GAS singeing and their causes

A. Incomplete Singeing

The most common causes of incomplete singeing are as follows:

1. Too low flame intensity
2. Too fast fabric speed
3. Too far distance between the fabric and the burner
4. Inappropriate (i.e. less severe) singeing position
5. Too much moisture in the fabric incoming for singeing.

If the fabric incoming for singeing has too much moisture in it, a significant amount of thermal energy will be used up in evaporating the fabric moisture rather than burning the protruding fibres, resulting in incomplete singeing.

B. Uneven Singeing Across the Fabric Width

The most common causes of widthways uneven singeing are as follows:

1. Non-uniform moisture content across the fabric width
2. Non-uniform flame intensity (uneven flame height) across the fabric width
3. Uneven distance between the burner and the fabric

   This may be due to misalignment or improper setting of the guide rollers

4. Uneven smoke evacuation over the burners

Common problems in GAS singeing and their causes

C. Uneven Singeing Along the Fabric Length

1.The most common causes of lengthways uneven singeing are as follows:
2.Non-uniform moisture content along the fabric length
3.Non-uniform flame intensity along the fabric length
- Variation in gas-air mixture supply
- Increasing or decreasing thermal energy of the flames during production

4.Change in fabric speed during singeing
5.Change in the distance between the fabric and the burner along the length

D. Horizontal Singeing Stripes

The most common causes of horizontal singeing stripes are as follows:
1.Rollers with an un-centred rolling action
2.Sudden fabric tension increase

E. Vertical Singeing Stripes

This may be caused by:
1.Total or partial blockage of flame outlet

Common problems in GAS singeing and their causes

F. Over-singing or Thermal Damage of the Fabric

The most common causes of over-singeing or thermal damage of the fabric are as follows:

1. Too high flame intensity
2. Too slow fabric speed or too long contact time between fabric and flame
3. Too close distance between the fabric and the burner or too deep penetration of the singeing flame into the fabric
4. Inappropriate (i.e. too severe) singeing position

G. Formation of Small Beads of Molten Material

This may be cause by:
1.Insufficient energy supply, when the thermal energy is not supplied quickly enough to be able to ignite the thermoplastic fibre rather than melting it.

Plate-Singeing m/c

- The main parts of plate singeing m/c :-
- i) Two copper plate &
- ii) Fire clay.
- ➢ Copper plates are set up on the fire clay which is heated with farness(gas and oil or oil jet).
- ➢ At first the fabric opening with guide roller is passed over the plate with drawn roller.
- ➢ When the plate was red heated with farness, copper plates drawn with draw roller which speed 135-225 m/min.
- ➢ Projecting fibres is burnt contact with heated plate.
- ➢ In this process, one side of fabric is singed

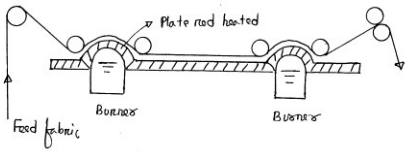


Fig: Plate singeing m/c.

❑ **Advantages:**

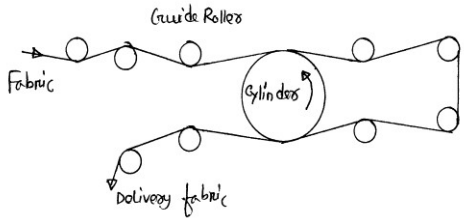
- Very suitable for back filter in finishing process.
- A certain amount of lusture is produce due to friction betⁿ the fabric & hot surface of curved plate.
- Improved lusture.
- Uniform singeing.
- More effective in case of valvet & pile fabric.
-

❑ **Disadvantage:**

- Very difficult to maintain the proper heat control of the plate.
- Only on side of the fabric is singed.
- Discontinuous process so more time taken.
- More labour cost.
- Not uniform singeing in stitch portion.
- Strength decrease

Roller singeing m/c

- This is more improved process than plate singeing m/c process. In this process, the fabric is singed one side also by heat but not red heat. If the fabric is singed by red heated roller, the heat is not even for this the result is not good



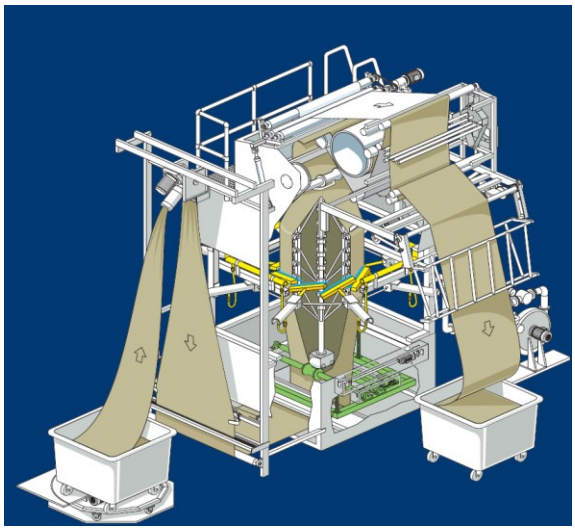
❑ **Advantage**:

- The surface temp of cylinder is more uniform than plate singeing m/c.
- This method is very suitable for valvet & pile fabrics.
- Uniform singeing.
- Lustre increase.

❑ **Disadvantage:**

- Only one side of the fabric is singed.
- Due to over use, small tinny groove created on the cylinder.
- Local cooling may arise on cylinder due to contact bet^{n} cold fabric & cylinder surface.

Circular singeing machine for tubular fabric



Features of knit Circular singeing machine

1. circular expander
2. enables tubular knit fabrics of cotton and its blends in all the usual diameters to be singed evenly and free of selvedge marks.
3. The fabric is guided onto the circular expander via a turntable and untwister
4. eight swiveling burners.
5. The diameter of the circular expander is power-adjusted and coupled to the burner adjustment system. Thus the gap between the burner and fabric remains the same
6. LCD monitor is provided to enable the data records

TECHNICAL DATA

- Circular expander – Diameter range 250 – 1200 mm (9.8" – 47.2")
- Circumference range 785 – 3770 mm (30.9" – 148.3")
- Singeing speed 40 – 120 m/min
- Fuels: Natural gas, butane, propane, LPG

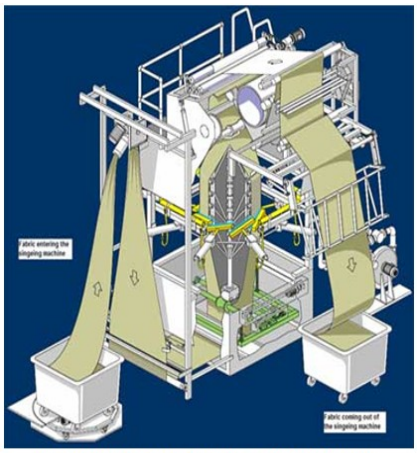


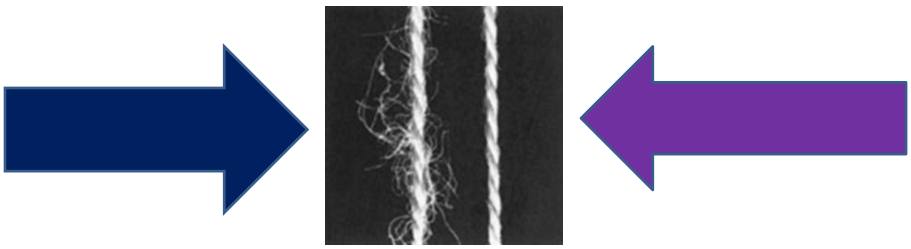
Fig. 8. Technical drawing of DORNIER singeing machine type EcoSinge for circular knitted fabric



Fig. 9. Singeing machine type EcoSinge in operation

YARN SINGEING

- Singeing is also done at yarn stage. For the yarn singeing it is an operation carried out in order to eliminate yarn hairiness.
- The singeing system consists of a package to package winder and a gas burner. The yarn is passed through the flame, which singes the protruding fibres that cause the hairiness. It runs at a rate of 400 to 1000 m/min. The machine must, in order to obtain even singeing, maintain a constant machine speed and an even flame.



Yarn Singeing Machine

Various parts of singeing machine

- Package holder cone form
- Handle of spindle
- Drum or roller
- Roller or drum plate
- Yarn guide
- Gas burner
- Feed Package holder
- Overhead cleaner
- Every m/c contains 40 package
- Each package weight is 2.5 kg
- Spindle plate (Tension box + sensor + yarn guide)

Machine Parameter

Yarn quality	Speed (rpm)
58/2, 60/2	1000
40/2, 20/2, 30/2, 24/1	850-900
Air pressure	15 bar
Gas pressure	17 bar

Bio-polishing

This is a process to remove the protruding fibers from the surface of a fabric or yarn through the action of an enzyme.

Importance of Bio-polishing:

- ✓ Bio-polishing is a finishing process that improves fabric quality by mainly reducing fuzziness from the fabric surface.
- ✓ Bio-polishing is a important process to eliminate micro fibrils of cotton through the action of enzyme.
- ✓ Cleaner surface of fabric is possible to achieve.

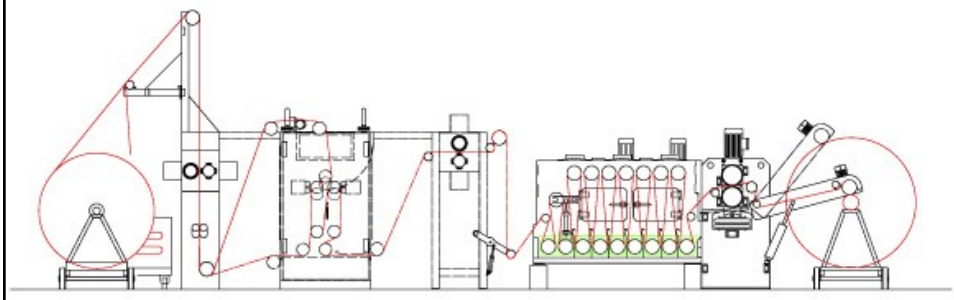
Optimum Condition of Bio-polishing

In bio-polishing, pH of the bath is adjusted within 4.5-5.5. Temperature needs to be maintained between 40-50 C and process time is maintained between 45-55 minutes.

Tentative Recipe for Bio-polishing

- ➢ Cellulase enzyme: 1%
- ➢ Acetic acid : 0.5 gm/liter
- ➢ pH : 4.5-5.5
- ➢ Temperature : 40-50*c
- ➢ Time : 45-55 minutes

CHAP:5
DESIZING



Desizing machine



Defination

- Desizing is done in order to remove the size from the warp yarns of the woven fabrics. Warp yarns are coated with sizing agents prior to weaving in order to reduce their frictional properties, decrease yarn breakages on the loom and improve weaving productivity by increasing weft insertion speeds. The sizing material present on the warp yarns can act as a resist towards dyes and chemicals in textile wet processing. It must, therefore, be removed before any subsequent wet processing of the fabric.
- Desizing is the first wet processing textile finishing technology employed to remove the sizing material from the fabric.
- It depends on
  - The solubility of the film forming polymer.
  - On the effects of numerous subsequent wet processing steps.
  - On the interactions with added chemicals.

Objects of Desizing

1. To remove the starch material from the fabric.
2. To increase the absorbency power of the fabric.
3. To increase the affinity of the fabric to the dry chemicals.
4. To make the fabric suitable for the next process.
5. To increase the luster of the fabric increase of dyeing and printing.

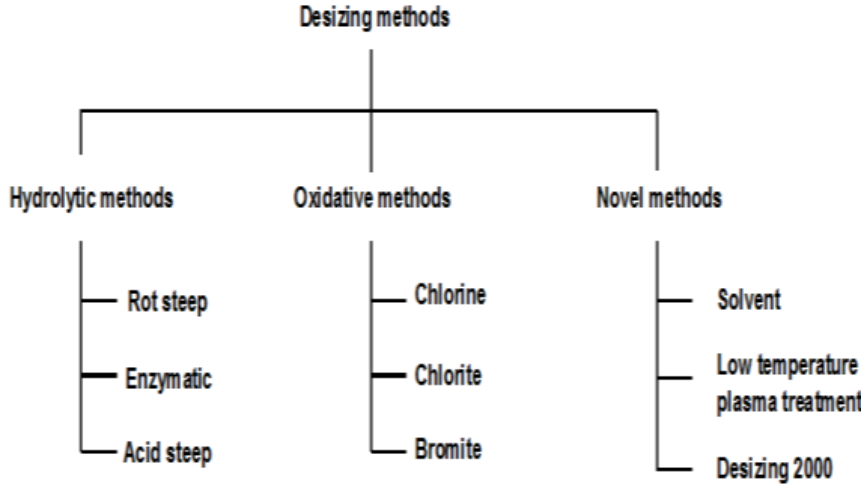
Mechanism

1. The object is to remove from the grey fabric the size that has been applied during weaving and thus to make the fabric ready for further processes.
2. The main ingredient in size that is not w ater-soluble is usually starch.
3. Chemically starch is poly-glucopyranose in w hich straight chain and branched chain polymers are present.
4. Both the constituents of starch are insoluble in water but they can be made soluble by hydrolysis of these long chain compounds to shorter ones.
5. Grey cotton fabric contains both natural impurities as well as 'added matter'.
6. The added matter is called 'size'. It is added by man in a process called 'sizing', as it facilitates weaving.
7. The size contains substances such as starch, thin boiling starch, CMC, PVA, vegetable oil, mutton tallow, etc.

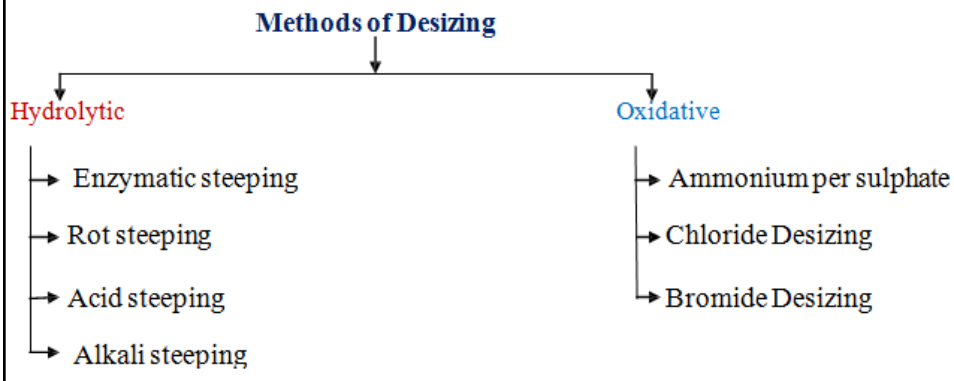
Factors of Size Removal Efficiency

- Type and amount of size applied
- Viscosity of the size in solution
- Ease of dissolution of the size film on the yarn
- Nature and the amount of the plasticizers
- Fabric construction
- Method of desizing, and
- Method of washing-off

Methods Of Desizing



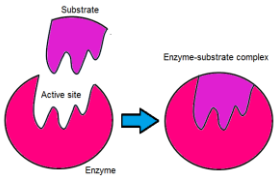
CLASSIFICATION OF DESIZING PROCESS



ENZYMATIC DESIZING

Enzymatic desizing is the most widely used method for the removal of starch, amylases being particularly suitable. The advantage in the use of enzymes is that starches are decomposed without damaging cellulose fibre. These are fairly sensitive to temperature changes from the optimum. Bacterial desizing agents like Rapidase are active over a wider temperature range and have certain other advantages, like tolerance of variation in pH.

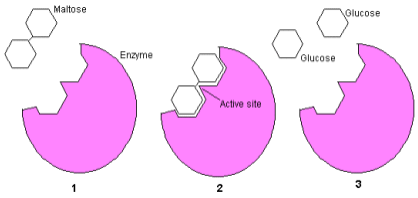
Enzymes suffer from one disadvantage that if the conditions of temperature and pH are not favourable, their desizing activity is destroyed. For example, their activity is destroyed they are deactivated above 75°C. An outstanding feature of enzyme desizing is the specific nature of the enzyme action. Thus diastase hydrolyses starch but does not tender cellulose. Therefore enzyme desizing is safer than acid desizing, where cellulose may also get hydrolysed if the concentration of the acid is higher than the optimum value.



Theory

We use Enzymatic Desizing. Enzyme are complex organic soluble by catalyst formed by living organisms that catalyze chemical reaction in biological process.

Amylases are enzymes that hydrolyse and reduce the moleculer weight of amylase and amylopectin molecule in starch. The traditional approach is enzymatic Desizing in which α -amylase or diastase enzyme is used to attack the 1:4 glycosidic links in the starch breaking down the macromolecules into small soluble saccharides such as maltose and glucose. Enzymatic Desizing is safer than acid Desizing where cellulose may also get hydrolysed if the concentration of acid is higher than the optimum value.



CLASSIFICATION OF ENZYMES

Mainly two types of enzymes. Such as:

1. **Animal enzymes:** Example: Viveral, Novofermosol, Degomma, Waste pancreas, Clotted blood, Liver, etc.

2. **Vegetable enzymes:**

There are two types vegetable enzymes.

a) Malt extract enzymes: Example: Diastafor, Diastase, Gabahit, Maltoferment, Maltostase etc.

b) Bacterial enzymes: Example: Rapidase, Biolase, Arcy etc

- **Other ways to define**
- **Cellulose enzymes** (degrade cellulose and if more time then cellulose damage and convert glucose)
- **Amalysis enzymes:** Outstanding features are – only hydrolyse the starch but do not attack cellulose. So it is used for de sizing

Types of amalysis enzymes

- **Malt extracts /malt preparation enzymes:** Malt enzymes are obtained by extraction of freshly germinated barley corns. Concentration, temperature, p^H are very important for the efficiency of enzyme. If the temperature is raised to the upper limit, it become destroyed(not useable)Best result at ---60^0C and destroyed at 75^0C
- **Trade name**:
  - Disastafor.
  - Ferment D
  - Terhyd MD etc.
- **Bacterial extract enzymes:** The ferment is obtained from bacillus subtilis which grows on grains of rice and breaks down the starch into water soluble dextrins.i.e. fermented rice→ produce bacteria→Extract→grow enlarge→prepared.Best resultat---70-75^0C and destroyed at ---95-100^0C
- **Trade name:**
  - Rapidase.
  - Biolase.
  - Bactolase.
  - Gelatase
  - **Acry**
- **Pancreatic analysis enzymes:** Pancreatic (animal) enzymes are prepared from the pancrease of slaughtered animals. A characteristics feature of pancreatic amylases id that they form milky-emulsion in water and are capable of modifying starch, albumen, pectins and fatts.
- **Trade name:**
- Fermosol
- Novofermosol DS
- Degomma etc.
- Viveral

CONDITION OF ENZYMATIC DESIZING PROCESS

Enzyme	Conc. (g/l)	Tempt. (°C)	PH value
Malt extract	3-20	50-60	6-7.5
Pancreatic	1-3	50-60	6.5-7.5
Bacterial	0.5-1	60-70	5.5-7.5

ENZYMATIC DESIZING PROCESS

Four faces must be considered for a successful enzymatic desizing process.

1. Preparation of the desizing mixture: Agents should be added:

1. Water
2. Wetting agent
3. Salt
4. Acid/Alkali
5. Enzyme.

First, salt and wetting agent are added than enzyme.

2. Saturation: Fabrics containing starch as sizing materials are difficult to wet out. So, it is mandatory that the mass of fiber and size be saturated to approx. 100% wet pick up.

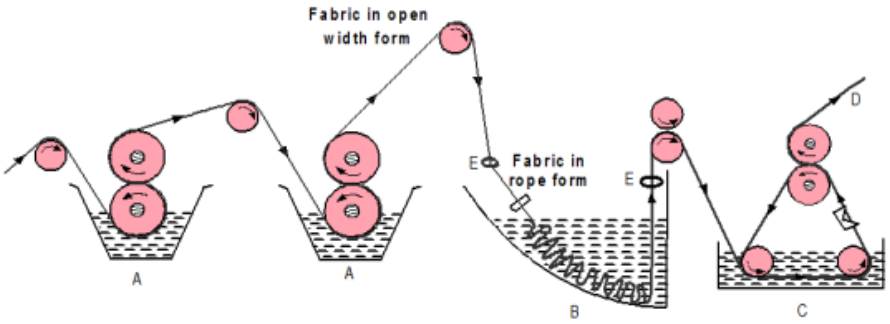
ENZYMATIC DESIZING PROCESS

3. Digestion: It means the process of converting starch to soluble materials. In a continuous process, fabrics are run through a steamer and conversion is accomplished during the steaming time available. In case of J-box, temperature range is 60 C to 90 C and time is 15 to 20 mins.

4. Washing: When desizing has been completed, it should be relatively easy to remove the short chain sugar as they are water soluble.
Main controlling points:

1. Temperature
2. PH
3. Fabric speed
4. Concentration

ENZYMATIC DESIZING



ENZYMATIC DESIZING

Advantages of enzyme desizing

1. Time required for the desizing process is less.
2. It is continuous process, so greater production can be achieved.
3. Closely constructed fabric can be easily desized, due to the effective enzyme action.
4. There is no chance for the cellulose to get hydrolysed, as in acid desizing.

Disadvantages

- If the conditions of temperature, pH and time are not properly maintained, the desizing activity of the enzymes is destroyed.

OXIDATIVE METHODS

The "oxidative desizing" technique mentioned above, is applicable not only for water insoluble sizing agents, but also for water-soluble ones. This technique is particularly useful for textile finishers, but also for water soluble ones. This technique is particularly useful for textile finishers dealing with many different types of fabrics and therefore sizing agents. In Oxidative desizing the starch is oxidized and this produces is also known as grey chemicking.

Desizing with Oxidizing agents

1. Though the use of oxidants for desizing of cotton fabric is widely accepted but their large scale industrial application is yet to be exploited.

2. The most important aspects of oxidizing agents are that they can be applicable to wide range of fabrics, the size content of which is often not known.

Table summarizes the necessary conditions for desizing starch in presence of some important oxidizing agents.

Desizing Conditions with Oxidising Agents

Oxidising agent	Process	Additives (pad-bath)	pH	Time (min)	Temp. (°C)
Hydrogen peroxide	Pad-steam	1-2 vol. H_2O_2, 7-15 g/l NaOH.	8-9	1-5	90
Sodium bromite	Pad-batch (cold)	1-3 g/l active Br_2, 20-30 g/l Caustic soda, 5-10 g/l Wetting agent.	7.5-8.5	15	20-40
Persulphate	Pad-steam	3-6 g/l Na-persulphate 8-10 g/l Caustic soda, 5-10 g/l Wetting agent.	10-10.5	1-3	95-100
Persulphate + H_2O_2	Cold-batch	40 ml/l H_2O_2 (25%) 10 g/l Persulphate, 10 ml/l Water glass, 10 ml/l NaOH 5 g/l Stabiliser, 5 g/l Wetting agent.	10-10.5	6-20	20-40

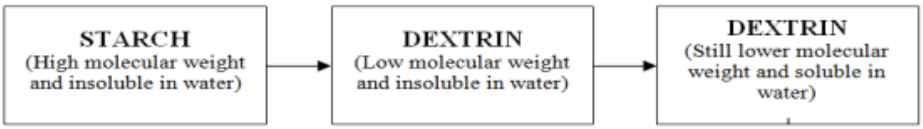
BROMITE DESIZING

Sodium bromite, the salt of bromous acid, HBrO2 (like sodium chlorite, the salt of cholorous acid, HClO2), has powerful oxidizing action on starch. This is due to the combined effect of bromous acid, HBrO2 and hypobromous acid, HOBr. This is accompanied by the conversion of bromine dioxide into oxygen and bromine. Hydrolysis of bromine thus formed produces more hypobromous acid. Of the different modes of oxidation of starch, the following one is the most likely one, involving breaking up of relatively stable either linkage of the glucose ring by sodium bromite.

If as shown above ethers are vulnerable to oxidation by sodium bromite, one would expect depolymerisation of the ether linkages (Oxygen Bridge joining two glucose units). Any significant degree of depolymerisation would then convert starch into water soluble products.

MECHANISM

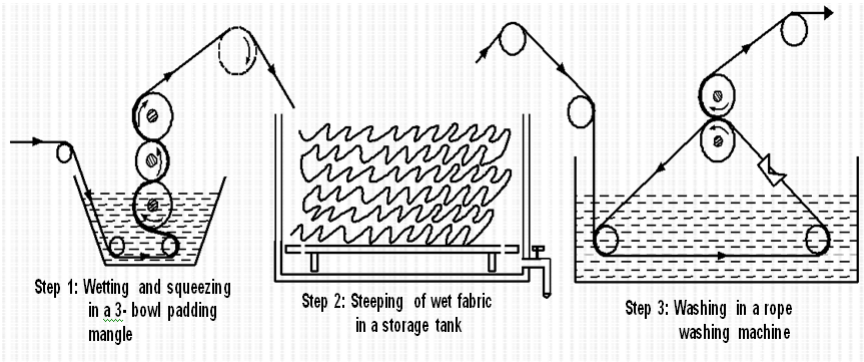
1. The main ingredient in size that is not water-soluble is usually starch.
2. Chemically starch is poly-glucopyranose in which straight chain and branched chain polymers are present.
3. Both the constituents of starch are insoluble in water but they can be made soluble by hydrolysis of these long chain compounds to shorter ones.
4. Thus, under suitable conditions, the following steps show the progressive hydrolysis of starch.
5. However, in desizing, the hydrolysis of starch is carried out only up to the soluble dextrin stage, as this can be removed off the desized fabric by means of an aqueous wash.



Rot Steep

1. This is the oldest and cheapest method of desizing.
2. Here no special chemical is used.
3. The cloth is first passed through warm water at 40C in a padding mangle where the cloth is squeezed to about 100% expression.
4. The cloth is then allowed to stand for 24 hours.
5. The microorganisms, naturally present in w ater, multiply and secrete starch-liquefying (hydrolysing) enzymes, which break down the starch present in the size to w ater-soluble products.
6. The cloth is then w ashed to remove these products.

Rot Steep



Rot Steep

Advantages

1. Rot steeping is the cheapest of all the desizing methods.
2. No chemicals are required.

Disadvantages

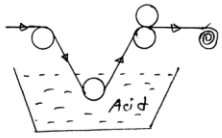
A large floor space is required for this process.

The process is slow, so desizing time is long.

Mildew may attack the cloth during steeping and cause stains on the fabric.

Acid Desizing

- Dilute sulphuric acid or hydrochloric acid may be used to hydrolyse the starch from the sized fabric.
- A 0.25% - 0.5 % solution of the acid at room temperature (30o C) is suitable for this process.
- The cloth is impregnated with the dilute acid solution in a two-bowl or three-bowl padding mangle and then stored for 8-12 hours in a closed concrete pit.



Fig; Acid steeping.

Acid Desizing

Advantages of acid desizing

1. Acid desizing is an economical process.
2. The process is effective and gives fairly uniform desizing, as it is a chemical-based process. It does not require specific conditions of pH and can be done at room temperature.
3. It is a much quicker process than rot steep desizing.

Disadvantage of acid desizing

1. The main disadvantage of the process is that mineral acid is harmful to cellulose fibres if proper care is not taken.
2. Especially during the storage stage, the acid-wet fabric must not be allowed to dry.
3. This would cause the formation of hydrocellulose, which will weaken the fibre.

Alkali desizng

- In this method the starch is removed by the alkaline hydrolysis. The fabric is treated with 0.4-0.6% Caustic soda solution at 60˚C to 70˚C and stored for 8 to 10 hours.
  Precaution: Care must be taken that, goods do not dry up, otherwise, it causes partial concentration of alkali.

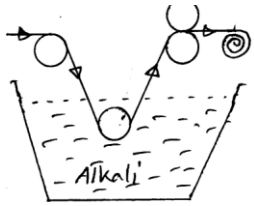


Fig: Alkali steeping

Alkali desizng

- **Advantage:**
- Economically use and cheap.
- Mercerizing can be done in same alkali (reusable).

- **Dis-advantage**:
- Considerable shrinkage may occur.

DESIZING EFFICIENCY TEST

Desizing Efficiency Test:
Desizing efficiency is found in two ways conventional and TEGEWA method.

Conventional Method:

In this method we first take the weight of the sized fabric, let it be W1. Then desize the fabric, dry & take the weight, let it be W2. After that the fabric is treated with 3gpl (35%) HCl at 700 C for 30 min. dry & take the weight of the fabric. Let it be W3.
Total size = W1-W3.
Residual size = W2-W3.

Desizing Efficiency = (Total size – Residual size)/Total size X 100.

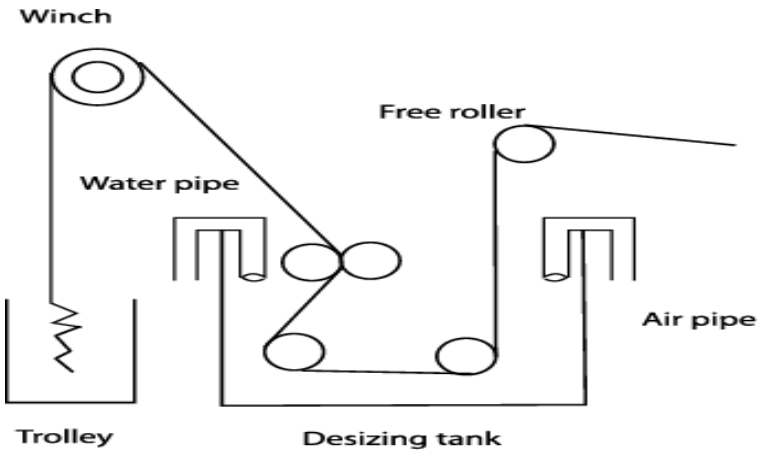
TEGEWA RATING

Reagent: potassium iodide (10 gm. Of KI (100%) in 100 ml water, add 0.6358 gm of iodine (100%) stir and shake; iodine is completely dissolved. Fill up to 800 ml with water then complete to 1000 ml with ethanol. (Shelf life approx 6 months only).
Method:

- Spot drop wise solution onto fabric.
- Rub in gently.
- Assess change of color.

Note: the test must be carried on fabric cooled down to room temperature; residual alkalinity has to be neutralities prior to the test.
Assessment: Grey fabric:
No change of color = no starch size present.
Pale blue to bluish = presence of starch size or blend
Violet =of starch size with synthetic size
Desized fabric:
Pale blue to bluish violet = refer to violet scale TEGEWA This indicates residual Starch content.

Desizing machine



Objects:-

- To observe the machine.
- To learn about the controlling points.
- To know about the mechanism.
- Overall maintenance of tank.

Main parts:-

- Desizing tank,
- Free roller,
- Winch roller,
- Tension roller,
- Squeezing roller.
- Liquor height
- If enzyme used Should control Ph(6-6.5) and temperature (55)

Controlling points of the machine

The machine should be maintained in several points of view. These points are mentioned on the next.

Squeezing roller:- The upper roller of the squeezing roller is soft and that of the lower roller is hard. The speed of the transmission of the cloth through the squeezing roller should be adjusted according to the quality of the fabric.

Roller movement:-Different free and guide roller should be moved easily. We have to look about that.

Water, steam line:- proper flow of water and steam should be maintained.

Proper fabric path:- In case of desizing proper path should be followed by the different roller.

Speed 80-100 m/min

Proper temperature:- In case of enzymatic agent the proper temperature should be maintained in the bath. (55)

Proper Ph: In case of enzymatic agent the proper Ph should be maintained in the bath. (6-6.5)

Liquor height: Should keep at standard level

Machine Facilities

Machine condition:

1. Mechanical defects
2. Rollers defects.
3. Utility supply pipes condition.
4. Electrical motor condition.

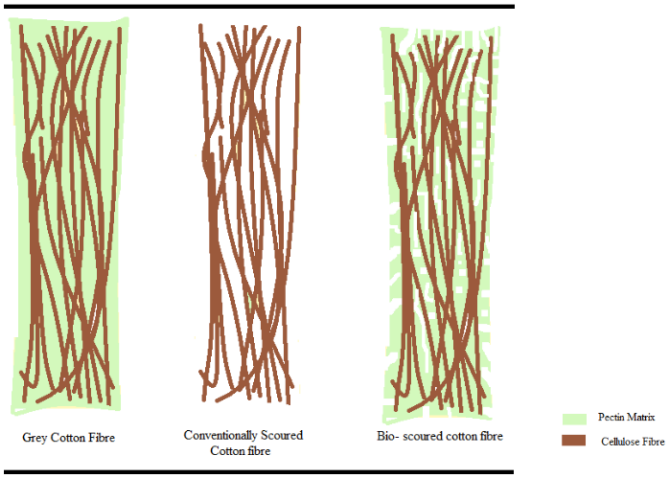
Machine Parameter:

1. Capacity of trough: 750 lit.
2. Nip pressure: 0.8 bar

Utilities:

1. Water,
2. Steam,
3. Electricity
4. & Gas :- 1=1.02kg/cm2

Chap:6
SCOURING



SCOURING

- **Scouring**: Scouring is the process by which all natural and additive impurities such as oil, wax, fat, hand dust etc. are removed to produce hydrophilic and clean textile material. It is one of the vital processes of wet processing.

- **Objects of Scouring:**
  - To make the fabric highly hydrophilic.
  - To remove impurities such as oils, waxes, gum, husks as nearly as possible.
  - To increase absorbency of fabric or textile materials without physical and chemical damage.
  - To produce a clean material by adding alkali.
  - To make the fabric ready for next process.
  - To remove non-cellulosic substance in case of cotton.

Theory of Scouring

Natural fibers contain oils, fats, waxes, minerals, leafy matter and motes as impurities that interfere with dyeing and finishing. Synthetic fibers contain producer spin finishes, coning oils and/or knitting oils. Mill grease used t o lubricate processing equipment mill dirt, temporary fabric markings and the like may contaminate fabrics as they are being produced. The process of removing these impurities is called Scouring.

Even though these impurities are not soluble in water, they can be removed by Extraction, dissolving the impurities in organic solvents, Emulsification, forming stable suspensions of the impurities in water and Saponification, Converting the contaminates into water

Mechanism

Saponification:

- The vegetable oil, which is immiscible with water, is glyceride of fatty acids. When such oils are heated with a solution of sodium hydroxide in water, the oil splits up into its constituents-fatty acid and glycerine. Glycerine is miscible with water easily and the fatty acids reacts with sodium hydroxide present in the solution forming its sodium salt i.e.soap which is also soluble in water. Thus oil is removed.

Emulsification:

- Wax and non saponifiable oils are removed by emulsification as they are immiscible in water. Normal washing soap is used as a emulsifying agent which makes emulsion of them.

Procedure of Scouring Process

1. Saponification of fats into water-soluble soap and water-miscible glycerin under alkaline conditions,
2. Hydrolysis of proteins into water-soluble degradation products,
3. Dissolution of hydrolysis to ammonia of simpler amino compounds,
4. Conversion of pectose and pectin into their soluble salts,
5. Dissolution of mineral matter,
6. Emulsification of unsaponifiable oils and waxes, and
7. Removal of dirt particles from the kier liquor by the detergent present therein

Scouring agents

Scouring Agent

Alkaline Agents
NaOH, KOH, Na_2CO_3, Liquid NH_3.
(Sodium metasilicate, Sodium Silicate, Sodium phosphate, Trisodium phosphate, Tetrasodium phosphate, Sodium tri-polyphosphate, borax etc.)

Surfactants
Anionic activator, Non-ionic activator

Emulsion Scouring

Organic Solvent

Chlorine System
Carbontetrachloride, Trichloroethylene, Perchloroethylene, Methylchloroform, Trichloromethane, Fluorine.

Hydrocarbon System
Benzene, Industrial gasoline, White spirit, Solvent naptha.

Chemical Used and Purpose

Chemical	Use
Caustic (NaOH)	Neutralize acidic materials, saponify glycerides (Waxes and Oils), and solubilize silicate.
Sodium Silicate	Penetrate and break down lignins in motes.
Surfactant	Reduces surface tension and minimize interfacial tensions.
Detergent	Emulsify oils, fats, and waxes; remove oil – borne stains; suspend materials after they have been removed.
Chelating (Sequestering) agent	Deactivate metal ions.
Builder (Salt)	Cause detergents to become increasingly effective.
Solvent	Assist emulsification by dissolving oily materials.

Sequestering of metal ions

- Scum formation because of Ca^{+2}, Mg^{+2}, Fe^{+3} metal ions
- Sequestering agents or chelating agents are negatively charged and are capable of forming strong ring structure with the metal ions present in hard water and in pectins.
- This prevents film and scum formation; precipitation of hard water. Advantages are better levelness and more brilliance in dyeing process, lower peroxide consumption, high degree of whiteness, and no catalytic damage during peroxide bleaching.
- Mostly used sequestering agent is EDTA.

Scouring Types

Form of Scouring:

1. Yarn scouring:
2. Hank form
3. Package form
4. Continuous sheet warp form.

Fabric scouring:

- Open width form …
- a. Jigger b. Pad batch c. Progressive jig
- Rope form
- a. Kier and b. Washer.

Scouring process:

- Batch process.
- Semi – continuous process.
- Discontinuous process.
- Modern process.

Scouring methods of Cotton

- Generally, there are two principle methods of cotton scouring.

1. Kier boiling process
    1. Horizontal
    2. Vertical.
2. Scouring in J or L box (Continuous process)

Scouring process depends on

Scouring process depends on: -

1. The type of cotton.
2. The color of cotton.
3. The cleanliness of cotton.
4. The twist and count of the yarn.
5. The construction of the fabric.

Kier boiling process/ discontinuous process

- Kier boiling process of cotton is performed in a kier boiler and the fabric is scoured in rope form, by alkali liquor.
- Kier is a large cylindrical iron vessel. The kier may be open one as closed one and horizontal or vertical.

1. **Open kier:** Open kier is not an air tight kier and scouring temperature 100˚C for 24 hrs.
2. **Closed kier:** Closed kier is air tight. Scouring temperature is 120-130˚C for 6 hours.

- **Kier capacity**: 200 kg – 5000 kg. but in Industrially 2000 kg kiering is very popular.

Standard Recipe

The amount of different chemicals used in a kier boiling may vary according to

- Hardness of water.
- Quality / Quantity of cotton.
- Fabric construction/ texture, yarn twist

Recipe

- Caustic Soda (Strong)→ 2-5% owf
- Soda ash (mild)→ 0.5-1.0%
- Sequestering agent→0.2-0.5%
- Wetting agent+Detergent→ 0.5-1.0%
- M:L→1:5 to1:7
- Temp→100˚C-120˚C
- Time→8hr(100^0C) - 2hr(120˚C)

Characteristic of boiler

- Heating is done by using high pressure steam, generated in an independent boiler instead of heating by direct boiler.
- Iron and steel take the place of wood as the mtl of construction of kier.
- Improvement in the handling of the goods by the use of automatic pilling device.
- Improvement of the circulation of kier liquor through the mtl by the use of powerful pump.
- Use of high pressure kier shorten the duration of boiling off and hence saves the cost of the steam, power etc.

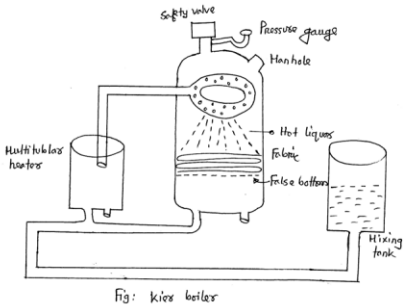
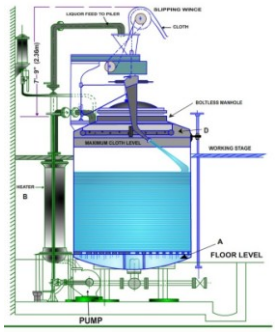


Fig: kier boiler



Working procedure

- Kier boiler is provided two tubes. One is at the bottom and another is top.
- There is one manhole for loading and unloading of sample.
- Preheater is also provided at the middle of these two tubes
- The liquor is prepared into the mixing tank by above chemicals then it brought into the pre heater and heated by the steam.
- The fabric is loaded in the m/c by man hole and kept in rope form.
- Then the hot liquor is pumped and sprayed by spader plate onto the fabric which is packed into the kier.
- The temperature of the liquor is about 100˚C and boiling for 8hrs.The excess liquor passes slowly over the packed cloth and percolates through the false bottom of the kier.
- Then this excess liquor is collect at the bottom of the kier and pumped into the auxiliary heater by a centrifugal pump and this cycle is repeat.
- Below the false bottom the liquor is free from the oxygen of air.
- After scouring ,the fabric is washed with 80^0C water otherwise impurities on the fabric would not be removed.
- Then the fabric is neutralizing with 0.1%-0.055 acetic acid. And then cold wash.

Kier boiler

Precaution

- Kier should be clean.
- Material should be packed evenly.
- No air pocket should be formed.
- The fabric should be immersed in liquor completely.
- After boiling the liquor should be removed in absence of water.
- Before starting, all the joining parts should be checked.
- The joint parts should be leak proof.
- Before scouring , the fabric should be starch-free.

Advantages:

- Material scoured well.
- Remove most of the natural colour and another adventitious.

Disadvantages:

- Failure of pressure gauge, boiler may be burst.

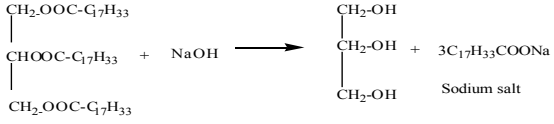
Continuous scouring process of cotton (Scouring in J- box)

- **Continuous scouring process of cotton (Scouring in J- box)**The scouring vessel is looks like the English letter 'J' hence, this process is called j box process. In the process, desizing, scouring and bleaching can be performed at a time.

Standard recipe	
Alkali (NaOH)	4-5gm/L
Wetting agent +Detergent	4-5gm/L
M:L	1:3
Pick up	90-100%
Impregnation Temp	70-80^{0}C
Impregnation Time	45-90sec
Storing time in J-box	2-4hr
Temperature in J-box	102-105°C

Process

- **Saturation:** Saturation is prepared by above recipe without caustic soda in the saturator. Then the wet fabric is passed through the guide roller and immersed into the solution by immersion roller. The fabric is saturated either by open width or in rope form. Here temperature is kept 70˚C – 80˚C for about 40-90 sec then the fabric squeezed and passed to the preheater.
- **Pre heater:** In pre heater, material is passed into the thermostatic controlling system at temperature 110˚C – 120˚C for 30 se and passed to the J- box by drawing roller.
- **J-box:** The fabric brought in J- box after pre heater. In j-box, solution of caustic soda are kept and fabric is stored in this solution for about 30 min, here temperature 100˚C. In j-box, NaOH, reacted with the impurities present in the fabric and finally removed.In J-Box generally 12000- 15000 lb fabric can be scoured after J- box the fabric is squeezed and passed to the washing unit.



- **Washing unit:** The water soluble impurities or products that are left on the mtl are removed here. First the materials are washed in hot water then cold water and finally dried

Process

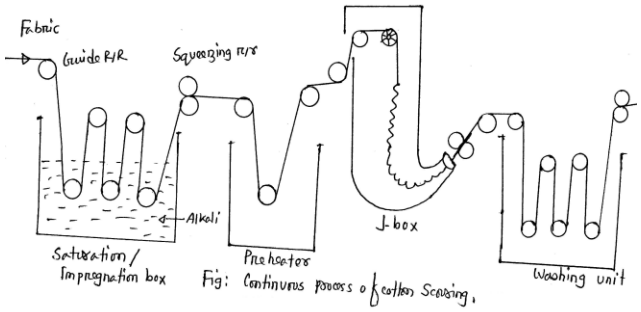
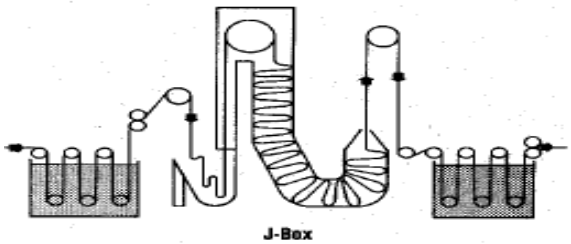


Fig: Continuous process of cotton Scouring.



J-Box

Continuous scouring process of cotton (Scouring in J- box)

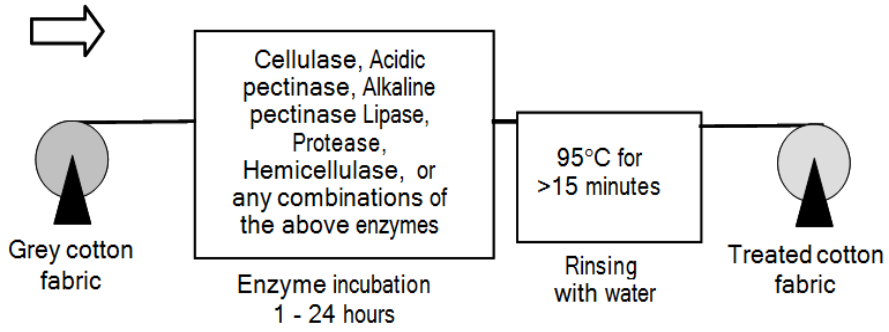
Advantage:

- This process is a continuous process, so it consumes less time.
- This process is economical use.
- Use for scouring obtained;
- Scouring, de sizing and bleaching performed at a time.

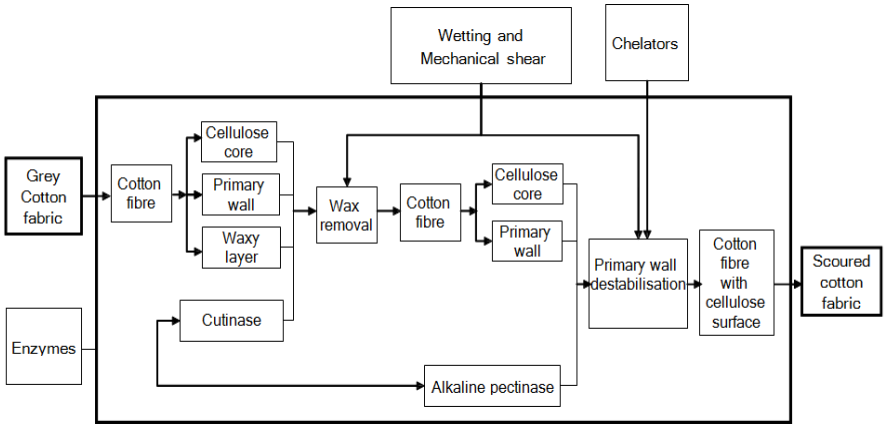
Disadvantages:

- The result is not good as compared with kier boiler.
- Huge damage may occur due to power failure.

Enzymatic Scouring process of cotton fabrics



Enzymatic Cotton Scouring process



Recipe Formulation

Enzymatic scouring was carried out by the following recipe-

- ✓ Enzyme : .....3 g/L or 0.5%
- ✓ Sequestering agent : .....1 g/L
- ✓ Wetting agent : .....2 g/L
- ✓ Emulsifier ..... 0.5-1.5 g/l
- ✓ Temperature : 60oC
- ✓ Time : .....30 min
- ✓ M: L : 1:10
- ✓ PH : 6-9

Scouring process of silk

Impurities present in silk:

- Sericin up to 30%
- The removing of above impurities in silk are called de-gumming.
- Agent used for degumming as – Soap, (Na_2CO_3 + $NaHCO_3$) Solution, synthetic detergent and pH of solution is 9-9.8

Recipe:

Soap	→0.5-0.75%
Sol^n of (Na_2CO_3 + $NaHCO_3$)	→3 galon /lb of silk.
Temp	→95˚C
Time	→30 min to 2 hrs.
pH	→10

Now a days synthetic detergents are used instead of soap for better performance. Noticed that pH of solution not beyond 10, other wise silk mtl may be hampered

The different types of silk on the basis of scouring

1. **Ecru silk:** Ecru silk is obtained by removing of 3-4% impurities (Sericin)

Soap solution	2-3%
Temp	Room temperature
Time	40-60 min

Used for warp yarn and for dark shade

2. Souple silk: Souple silk is obtained by removing of 10% impurities (Sericin)

Soap solution	10%
Temp	Room temperature
Time	1-2 hrs.

Used for medium shade

The different types of silk on the basis of scouring

3. Boiled off silk:
Sericin is removed up to 30% This process required two bath processes:

	1st Bath	2nd Bath
Soap solution	30%	10-15%
Temp	90-95^{0}C	95˚C
Time	60-90mins.	1-3 hrs
Soda ash		1-2%

Used for white and light shade

Scouring of wool

- Wool contain 30 – 60% grease and also contain suint, fats etc as impurities.
- Mild alkali Ammonium Carbonate or Sodium Carbonate is used in wool scouring because wool is degraded by caustic soda or even low strength alkali at high temperature.
- So, pH of liquor should be 8 – 10.5. As the melting point of fat in wool is 40-50˚C. Hence applied temperature should be 50-55˚C. Also the soap used in washing must be made of fatty acid of low melting point. Because they are more soluble at lower temperature.
- Synthetic detergent are more preferable instead of soap and salt are used in the solution.

Standard recipe

- The standard recipe for wool scouring used in Harrow m/c and Jet scouring m/c of different bowl are
- p^H= 10 or below 10
- Capacity of bowl= 1500 gal.
- Each bowl are rectangular in shape and made of cast iron. Wool scouring is performed in every bowl and squeezed. In 4th bowl, the mtls are washed in water and finally dried

Order of Bowl	Soap%	Na_2CO_3	Temp(^{0}C)	Time (min)
1st	0.8	0.2	49-52	2.5-3
2nd	0.9	-	46-49	2-2.5
3rd	0.37	-	43-46	2.0
4th	Water	-	40.5-43	1.5

Scouring of Jute: (Bast fibre)

- Bast fibre, jute, linen etc. are mulicellular fibre, having polygonal shape. The cells are cemented with each other by lignin which is non-cellulosic unit and is about 11-14%.
- If strong alkali is used for jute scouring, its strength is reduced as lignin is removed and cells are separated. So, mild alkali is used in jute scouring at low temp.
- Scouring is not required for Hessian and Sacking jute

Standard recipe

Materials	Quantity
Na_2CO_3	4-6gm
Wetting agent + Detergent	0.5-1gm/ltr
Water softening agent	0.5-1.0gm/ltr
M:L	1:15-1:20
Temp	Boiling(100^0C)
Time	2-4 hrs.

- Better performance is obtained, if 1% Oxalic acid solution is used to boil jute fibre before scouring.
- After scouring, jute is treated with CH_3COOH or in HCl

Solvent Scouring Process

There are commercial processes where textiles are cleaned with organic solvents. Fabrics processed this way are said to be "Dry Cleaned". Although not widely used as a fabric preparation step, it is an important way of removing certain difficult to remove impurities, where a small amount of residuals can cause downstream problems. Garment dry-cleaning is more prevalent.

For fabrics that do not have to be desized, solvent scouring is an effective way of removing fiber producer finishes, coning and knitting oils. Knitted fabrics made from nylon, polyester, acetate and acrylics, are particularly amenable to this method of preparation. Wool grease is effectively removed by solvent scouring. Solvent

Extractions are particularly useful in the laboratory for determining the amount of processing oils added to man-made fibers and the residual amounts of oils and waxes left by aqueous scouring. Properly controlled, fabrics can be produced with very little residual matter.

Blends

Scouring fabrics with a blend of fibers requires consideration of the sensitivities of each fiber to scouring chemicals and to process conditions. Sensitivities to consider when scouring blends are:

Cotton:	Resistant to strong alkali. Degraded by acid.
Rayon:	Sensitive to alkali. May be dissolved by hot alkali.
Wool:	Degraded by alkali.
Acetate:	Hydrolyzed by alkali.
Polyester:	Hydrolyses under extreme conditions of alkali and heat.

Synthetic Scouring

- Synthetic fibers generally do not contain naturally occurring impurities like natural fibers. However spin finishes, knitting weaving oils, antistatic agents are added to improve physical as well as mechanical properties. Other are dirt etc.
- For polyester weak and low concentration of alkalis are used at low temperature.
- Special precaution is necessary when polyester is scoured with strong alkali at higher temperatures and care has to be taken not to hydrolyze the fiber.
- Nylons are scoured with mild alkali and detergents. Generally non-ionic detergents are used in scouring of nylons.

Estimation of Scouring

Estimation of Scouring:

1. Determination of weight loss
2. Absorbency Test
   - Immersion Test
   - Drop Test
   - Spot Test
3. Column Test

AATCC Test Method Number 79

Properly scoured fabric should wet out faster and be more water absorbent. AATCC Test Method No. 79 is used to measure fabric wetting. A drop of water is placed on the fabric and the time it takes for the drop to penetrate the fabric is recorded. The faster the wetting time, the more absorbent the fabric.

Estimation of Scouring

Determination of Weight Loss:

- Standard weight loss is 4 – 8%
- If weight loss is less than 4%, it can be said that scouring was not well
- If weight loss is above 8% then it can be said that fabric damage has occurred.

$$\text{Weight loss\%} = \frac{Weight\ before\ scouring - Weight\ after\ scouring}{Weight\ before\ scouring} \text{ x } 100$$

Measurement of Weight Loss:

5 gm of dried samples is treated with 200 ml of 1% NaOH for 1 hour at 80° C after which sample is well rinsed and run out in hot water. It is then treated in 200 ml of 0.5% HCl at 80° C for 1 hour, after which sample is once again rinsed, boiled for 1/2 hour in distilled water, dried & weighted.

Estimation of Scouring

Immersion Test:

- Sample size is "1 cm x 1 cm"
- If the fabric floats on the water, then it may be said that the fabric is unscoured
- If the fabric is immersed within 5 seconds then it may be said that the fabric is scoured well.

Drop Test:

- Solution 0.1% Direct red
- If dye drop is absorbed within 1 second, then the scouring is of standard level
- If dye drop is absorbed within 0.5 – 0.8 second, then the scouring is of good level.

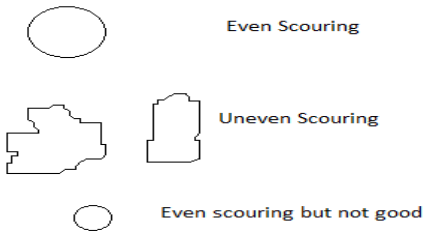
A drop is allowed to fall on the fabric by a pipette from 1 or 2 inch above fabric and time in second is measured until the dye drop is fully absorbed.

Estimation of Scouring

Spot Test:

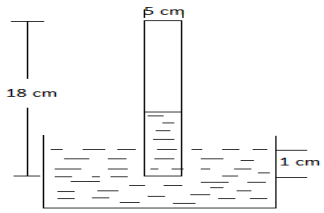
- Solution of 0.1% direct red
- Pour uniform drop on fabric

Size of droplet is observed



Column Test/Wicking Test:

- Solution of 0.1% direct red
- Sample size "5 cm x 18 cm"
- Observation time 5 minutes



Estimation of Scouring

Observation:

1. The length of fabric is observed in solution.
2. If the absorption rise up as high as 30 mm then it may be said that the fabric has good scouring
3. If the absorption rise up as high as 50 mm then it may be said that the fabric has excellent scouring.

Assessment of pretreatment by absorbency test:

Verdict	Spot Test	Wicking Test		Wicking rate		
		5 minutes	10 minutes	1 cm	2 cm	3 cm
Good Pretreatment	1 – 5 Second	30 – 50 mm	50 – 90 mm	3 – 5 sec	10 – 30 sec	40 – 70 sec
Poor Pretreatment	More than 10 seconds	Less than 30 mm	Less than 50 mm	More than 10 sec	More than 30 sec	More than 100 sec

Souring

The treatment/the process by which the fabric, after processing with alkali or scouring, is treated with Acetic Acid, Hydrochloric acid or dilute H_2SO_4 for removing alkali or neutralization of alkali is souring.

Scouring	Souring
1. To remove oil, waxes gum soluble impurities.	1. Not to remove any impurities, only for alkali neutralization.
2. Scouring is done in alkali solution.	2. Souring is done dilute HCl or H_2SO_4
3. Required heat to boiling.	3. No need of heat.
4. Need of definite time.	4. No need of definite time.

Chap:7
BLEACHING



DEFINATION

Bleaching

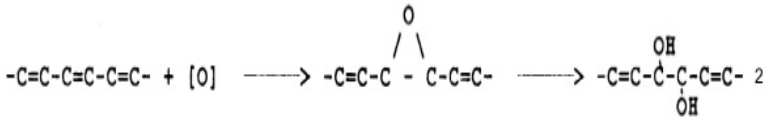
The process of decolorize the natural coloring matter present in the cloth treating with some oxidizing agent and reducing agent and ensure the permanent whiteness is called bleaching.

Objects

- A high uniform absorbency of fabric to water and dye stuffs.
- Uniform degree of whiteness.
- Fabric should not damage and DP should remain high.
- Destruction of natural coloring matters from the fabric.
- To ensure a level dyeing properties.
- To make the textile materials suitable for subsequent processing. (Dyeing, printing, etc.)

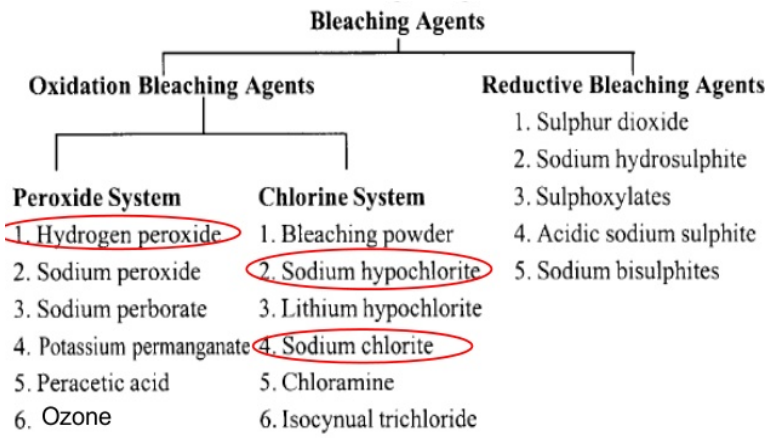
Mechanism of Bleaching

The mechanism of bleaching is very complicated and not completely understood. One opinion is that the color producing agents in natural fibers are often organic compounds containing conjugated double bonds. Decoloration can occur by breaking up the chromophore, most likely destroying one or more of the double bonds within the conjugated system. The bleaching agents either oxidize or reduce the coloring matter … thus whiteness obtained is of permanent nature. Primitive bleaching - expose scoured fabric to the sun - light served as an oxidation catalvst.



Bleaching Agent

A bleaching agent is a substance that can whiten or decolorize other substances.



Application of bleaching process to fiber types

Type of fiber	Alkali	Sodium chloride	Sodium hypo chloride	H_2O_2	Reducing agent
Cellulose	++	++	++	++	++
Animal	_	_	_	++	++
Polyamide	+	++	_	(+)	++
Polyester	(+)	++	+	+	+
Polyacrylonitrile	_	++	+	(+)	+

- ++ Process highly soluble,
- +, Process applicable without fiber damage but without achieving and whitening effect.
- (+) , Process applicable only with special precaution.
- _ process unsuitable

Bleaching auxiliaries

- **Wetting agents:** Sulphonated oils, fatty alcohol sulphates, fatty acid condensates
- **Activators** for bleaching with H2O2 is usually NaOH which controls the pH
- **Stabilizers:** Very important for the bleaching with hydrogen peroxide, suitable products are sodium silicate ,phosphates, organic complexing agents, etc.
- **Sequestering agents:** They help to sequester out metal ions such as EDTA.
- **Corrosions inhibitors** for sodium chlorite bleaching: fatty acids condensates, nitrates and phosphates.

Hypochlorite bleaching

- Sodium hypochlorite (NaOCl)or Calcium hypochlorite [$Ca(OCl)_2$] may be used as hypochlorite bleaching agent.
- When Calcium hypochlorite or Sodium hypochlorite is hydrolised, hypochlorus acid is formed which ionizes under a certain condition any give hypochlorus ions which are responsible for bleaching action. Alkaline condition favours the reaction-

  $Ca(OCl)_2+H_2O+CO_2 \rightarrow CaCO_3+ 2H0Cl$

  $HOCl \rightarrow H+ + Cl-$
- Hypochlorous ion (responsible for bleaching)&

  $NaOCl+H20$ NaOH + HOCl

  $HOCl \rightarrow H+ + Cl-$
- Hypochlorous ion (responsible for bleaching)

- When calcium hypochlorite is used, it reacts with atmospheric carbon dioxide to give calcium carbonate as white precipitate.
- $Ca(OCl)_2+H_2O+CO_2 \rightarrow CaCO_3 \downarrow + 2H0Cl$
- $CaCO_3$ deposited on the fabric causing harsh handling and uneven dyeing, hence it have to separate and souring(acid treatment) is done to remove it.

Differences between $Ca(OCl)_2$ and NaOCl bleaching

In textile hypochlorite bleaching sodium hypochlorite [NaOCl] or calcium hypochlorite [$Ca(OCl)_2$] may be used as hypochlorite bleaching agent.

$Ca(OCl)_2$	NaOCl
1.It is unstable	1.It is stable
2.It produces $CaCO_3$ precipitate	2. It doesn't produce any precipitate
3.It makes harsh feeling on the fabric	3.It doesn't make harsh feeling on the fabric
4.Comperatively cheaper than NaOCl bleaching	4.Higher cost than $Ca(OCl)_2$ bleaching

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Effect of P^H

The effect of **P^H** is very important during hypochlorite bleachig –

- When **P^H** 2 – 4: Produce more chlorine which toxic and corrosive.
- When **P^H -4:** More stable HOCl i.e. HOCl does not divided into H^+ and OCl^-
- When **P^H 7** neutral: Decomposition of NaOCl is very high. So it breaks cellulose. This point is called "Damage point".
- When **P^H** 7-8: Quick bleaching which tends to degradation of cotton fabric.
- When **P^H** 9.2 – 11: Fabric become neutral and more stable.
- When **P^H** 11 -13: It is the perfect range, but it require more time. If **P^H** =11, bleaching needs 130 hrs. and if **P^H** =13, bleaching needs 40 hrs. to be finished

- From the above chart it is clear that, bleaching by hypochlorite should be done in alkaline medium and not in between the value of **P^H** 2 -7
- Ideal P^H for Hypochlorite bleaching →9.2-11.0
- P^H is maintained at 10. For P^H maintain-
- 0.2-0.8% Caustic soda
- 2-3% Soda ash

Antichloro treatment

- In case of hypochlorite bleaching, Hypochlorus ion produce during bleaching. This (OCl^-) ion will react with residual protein into fibre and produced Chloramine (NCl) which is corrosive and unhygenic. After bleaching, the chloramine react with moisture and gradually cotton become yellowish due to forming of HCl.
- To remove >NCl, Antichlore treatment done.
- For the antichlore treatment of cellulosic fibre the general recipe is as follows:

➢ $NaHSO_4$ – 0.2 – 0.6%
➢ Temp. – Rooms
➢ Time – 10- 20 min.

TYPICAL BATCH PROCEDURE

a. Formulation:



b. Bleach Cycle:



Uses

1. Hypochlorite is used mainly to bleach cellulosic fabrics. It cannot be used on wool, polyamides (nylon), acrylics or polyurethanes (spandex).
2. These fibers will yellow from the formation of chloramides.
3. Bleaching with hypochlorite is performed in batch equipment.
4. It is not used in continuous operations because chlorine is liberated into the atmosphere.
5. Over time, the pad bath decreases in active chlorine causing non-uniform bleaching from beginning to end of the run.

Hydrogen Peroxide (H_2O_2) Bleaching

- Hydrogen peroxide was first used to bleach cotton in the 1920's. By 1940, 65% of all cotton fabrics were bleached with hydrogen peroxide, largely brought about by the invention of the J-box which lead to continuous processing.
- Today, it is estimated that 90 to 95 % of all cotton and cotton/synthetic blends are bleached with hydrogen peroxide.
- It is available commercially as 35, 50 and 70 % solutions.
- It is a corrosive, oxidizing agent which may cause combustion when allowed to dry out on oxidizable organic matter. Decomposition is accelerated by metal contamination and is accompanied by the liberation of heat and oxygen, which will support combustion and explosions in confined spaces. The material is an irritant to the skin and mucous membranes and dangerous to the eyes

Hydrogen Peroxide (H_2O_2) Bleaching

- Bleaching is done to remove natural color & impart to whiteness of the fabric.
- $H_2O_2 \rightarrow$ 0.5-6gm/L, In terms of Dark fabric 2-3 gm/L.
- As bleaching agent H_2O_2 is used.
- $H_2O_2 \rightarrow H^+ + HO_2^-$
- HO_2^- is responsible for whiteness of the fabric.
- HO_2^- stay unstable on the reaction. It tends to be O_2 form after certain time.
- To reduce O_2 formation stabilizer (Stager-B) is used in the bath.
- H_2O_2: Stabilizer ratio should be 6:1.
- If H_2O_2 is more on the bath then Stabilizer should give more on the bath.
- If stabilizer is not present in the dyeing factory then H_2O_2 supply should be more.

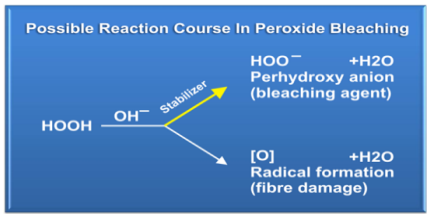
- If we don't use stabilizer O_2 formation will be increase or If we don't use per oxide killer then O_2 react with OH which come from cellulose bond, & form CHO by oxidation. By further oxidation CHO will form COOH that's why cellulose bond will be loose.
- $O_2 + OH \xrightarrow{OXIDATION} CHO \xrightarrow{OXIDATION} COOH$
- COOH is responsible for abrasion of the fabric.
- There are two solution of H_2O_2 bleaching agent (1) H_2O_2(35%), (2) H_2O_2(50%).
- If we need H_2O_2 (50%) in bath but have H_2O_2 (35%) then we have to supply more amount of H_2O_2 (35%) _Because of density of the solution.
- Peroxide killer(0.2-0.5gm/L, 20min) is used to control O_2 formation
- Peroxide killer depend on the amount of H_2O_2, if H_2O_2 is more than Per oxide killer should give more on the bath.
- We can control P^H by using CH_3COOH if P^H is basic.
- CH_3COOH is used for neutralization process.
- We can use CH_3COOH (1gm/L) to bring the fabric in base form to acid form.

Bleaching action of Hydrogen per oxide

- Under certain condition, particularly regard to **P^H**, hydrogen peroxide will liberate hydrogen ion and per hydroxyl ions in the following manner
- Per hydroxyl ions responsible for bleaching.
- Alkalinity favours the liberation of Per hydroxyl ions because the positively charged hydrogen ion is neutralized but excessive alkalinity cause the peroxide to become unstable. The hydro – peroxide ion is resposible for bleaching action.
- In presence of catalyst such as CaCO3, Fe, Cu, Cr, Mg etc. liberated oxygen by decomposing H_2O_2 and lower the strength of H_2O_2.

$$2H_2O_2 \longrightarrow 2H_2O + O_2$$

- Hence ($2^0 – 7^0$) or (6^0 - 8^0) hardness are suitable for bleaching.



Difference between H_2O_2 bleaching and Hypochlorite bleaching

H_2O_2 bleaching	Hypochlorite bleaching
1. Per hydroxyl ions(HO_2-) are responsible for bleaching.	1. Hypochlorous ions (OCl-) are responsible for bleaching.
2. Permanent and white are obtained	2. Permanent and white are not obtained as hydrogen per oxide bleaching
3. Temp. near to boiling i.e. above 90^{0}C	3. Room temp.
4. Can be performed in scouring.	4. No scouring action is done.
5. Universal bleaching agent.	5. Not Universal bleaching agent.
6. Can be bleached both cellulosic and protein fibre.	6. Only cellulosic fibre can bleach.
7. No need of antichloro treatment.	7. Need antichloro treatment.
8. Less possibility of fabric wastage.	8. More possibility of fabric wastage.

Standard or general or typical recipe

Chemicals	Bleaching	Single stage Scour – bleach
H_2O_2 (35%) OR H_2O_2 (50%)	5-6% 3-4%	5-6% 3-4%
Caustic soda	0.2-0.5%	4%
Soda ash	0.5-1.0%	3%
Wetting agent		0.5-1.0 %
Sodium silicate (Stabilizer)	0.5-1.5%	2.0-3.0%
PH	9-10.5	Same
M:L ratio	1:10	1:15
Time and temp.	98-100°C-4hrs 102-105°C-3hrs 110°C-1hr	100°C-4.5 hrshrs 102-105°C-3.5 hrs 110°C-1.5hrs

Function of required chemical

- **Stabilizer**: It makes complex compound with catalyst but not react and stop the oxygen generation into solution and preserved the strength loss of H_2O_2. Generally, Sodium silicate used as Stabilizer.
- **Catalyst:** The water used in bleaching may present Cu, Zn etc. which acts as catalyst and destroy H_2O_2 by generation oxygen but this oxygen have no bleaching power.
- **Alkali:** Without alkali HO_2^- producing is slower. On the other hand, huge alkali present in solution, H_2O_2 decompose and produce O_2. The function of alkali (Caustic Soda) is to maintain
  PH 9.2 – 11.5.

PH	Decomposition of % of H_2O_2
10.4	7
11.1	15.5
11.9	19.0
12.2	25.0
12.6	59.0

Function of required chemical

- **Soda ash:** To also maintain P^H and more whiteness.
- **Wetting agent:** To wet the fabric by lower interfacial tension.
- **Water:** For better action of sodium silicate, some magnesium salts are added hence 2^0 -7^0 hardness of water are used. If pure soft water are used, then 0.1 – 0.2 gm/L Magnesium sulphate.
- **Temperature:** Normally the temperature increase, the stability of H_2O_2 reduces.

  If temperature is 20^0c or less than 20^0c, H_2O_2 is more stable even alkaline condition. Not good bleaching below 80^0C temperature bleaching.
- **Impurities in cotton:** Higher the impurities, higher the stability of H_2O_2 and higher the bleaching acts as a stabilizer.

  As a result, H_2O_2 do not break and proper leaching is performed.

Advantages of H_2O_2 bleaching over other bleaching agent

- H_2O_2 does not react with residual protein of fibre and hence no need antichloro treatment.
- Permanent white cotton is obtainable and the bleached fabics are highly hydrophilic since the waxes are solublised and removal by the hot alkaline solution.
- Its reaction products are relatively non toxic and it decomposes to oxygen and water thus reducing greatly the effluent pollution of the bleaching plant.
- H_2O_2 bleaching is carried out in alkaline medium and elevated temp. is about 100^0c, hence scouring and bleaching completed together.
- Small amount of impurities present in cotton fibre, give stability of H_2O_2 in solution and so needs scouring. For this reason, impurities in cotton acts as stabilizer in H_2O_2 bleaching.
- Weight of fabric after H_2O_2 bleaching is higher than that of hypochlorite bleaching.
- Tensile strength is greater after H_2O_2 bleached fabric than that of hypochlorite bleached.
- Another advantage is degradation possibility of fabric is less due to over bleached.
- Hard water preferable.(2^0-7^0C).
- Bleaching and dyeing can be sometimes combined in a single operation.
- The no. of operation and stages in the bleaching can be reduced and continuous one stage process can be worked.
- It si compatible with the most fibres and can be applied to a wide variety of fabric under a wide range of bleaching condition and machines.

Disadvantages

- Bleaching is slow unless high temperature is applied energy
- Catalytic decomposition of H_20_2 occurs along with catalytic degradation of cellulose due to iron, Ne, Cu and Pb hydroxide present in the bleaching solution or I the fabric.
- The above metals and their alloys cannot be used as material of construction of H_20_2 bleaching containers

H_2O_2 universal bleaching agent

- Hydrogen peroxide is successfully used to bleach both cellulosic (vegetable) and protein (animal) fibre.
- In case of cellulosic fibre, H_2O_2 permanently destroy the natural color and obtained good result.
- In case of protein fibre H_2O_2 oxidized the protein mtl. but if no chloride ion. For this di-appearance, it has no effect on protein fibre and also destroys the natural color permanently.
- H_2O_2 bleaching is done at elevated temperature is about 1000c in alkali medium and hence scouring and bleaching can be performed together.
- So, H_2O_2 is called universal bleaching agent.

Methods of bleaching with Hydrogen Peroxide

- There are two chief methods for bleaching of cotton goods with H_2O_2.

1. Bleaching in kier (Discontinuous)
2. Bleaching in J – Box (Continuous)

Bleaching in kier

- The simplest cotton bleaching method with H_2O_2 is in kier. When the conventional kier is used, there is the risk of catalytically decomposition of the H_2O_2 by the rest is the kier. So sodium silicate (Stabilizer) and cement should be used to prevent the wastage of hydrogen peroxide. The circulation of liquor may be provided either by centrifugal pump or by a steam injector.

Chemicals	Amount
H_20_2 (35%)	2-4%
Caustic soda	0.2-1%
Soda ash	0.5-1.0%
Wetting agent	As required
Sodium silicate (Stabilizer)	2-3%
PH	9-10.5
M:L ratio	1:10-1:20

Working principle

- At first the chemicals are dissolved in a separate mixing tank, then it transfer to the kier to fill its one third.
- Materials are feed and liquor circulation is continue.
- When mtls are half filled of the kier, then steam is supplied and when loading completed, liquor are circulate for complete immersion of materials and raise the temp. from 40^0-60^0c.
- After that, the temp is raised at 65-70^0c. At this stage, flow of steam and liquor circulation turned off for 10 min. to remove air from kier.
- Again, raised the temp. at 80^0c and flow of steam and liquor circulation turned off for 10 minutes.
- After that, kier is completely closed and raised the temp. 110^0c and kept it 1-3hrs.
- If the kier is open then it kept for 5-6 hrs. thus bleaching is done.
- At 90^0c the process is slow. So the bleaching is done at 95-100^0c or 102 to 105^0c. But the temp. could be raised to maximum of 110^0c. If the temp. is above 110^0c the risk of degradation of cotton is very high.

Kier boiler

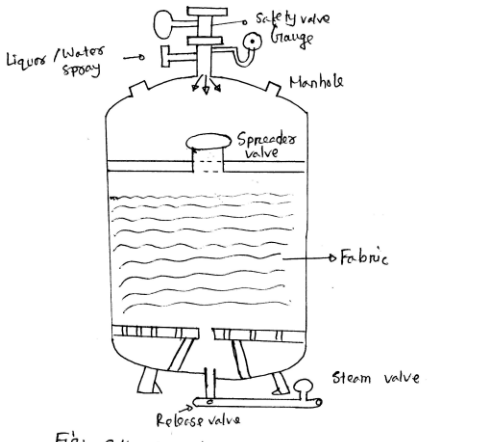
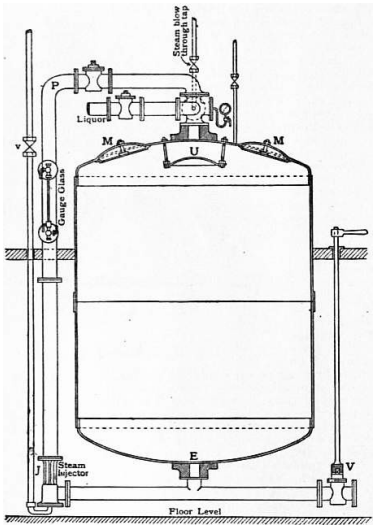


Fig: Cotton bleaching process in high pressure kier



Bleaching in j-box: (continuous bleaching process)

- The bleaching vessel is looks like 'J' the english letter 'J' hence this process is J – Box process. J-Box is made of stainless steel or ceramic. Scouring in J-Box mode, the scouring in short duration and easily controlled. As scouring and bleaching are performed continuously, hence two J-Boxes are required. In textile mill, this process are used commercially. Capacity of J-Boxes are 6-40 tones per day.
- Following recipe is required for bleaching

Caustic soda – 4-5 g/l

H_2O_2 (35%) – 4-5 g/l

Sequestering agent -1-2 g/l

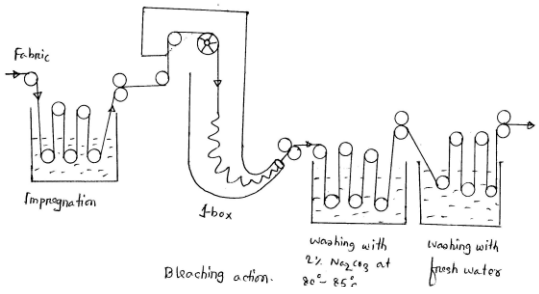
Wetting agent -1 g/l

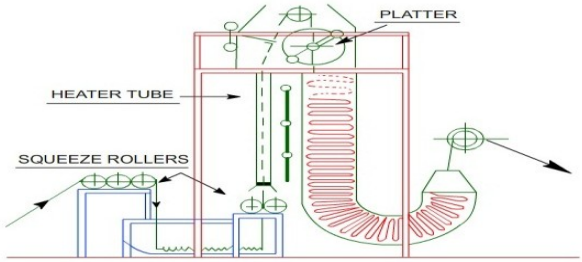
PH – 10.9

Working principle

- **Processing in impregnation box:** At first, above chemicals are dissolved and make liquor is impregnation box. After processing from 1st J-box for scouring the mtls. then wash. The mtls. are immersed into impregnation box by immersion roller then squeezed and brought to the J – box.
- **Storing in J-Box:** After impregnation of fabric from bleaching liquor then, the mtl. store in J-Box where, steaming is done at 93-99^{0}c for 60-90 mins. and bleaching is completed.
- **Washing:** Then the mtls are washed in hot water then in cold water and finally dried or first the mtls are washed with 2% solution of sodium carbonate at 80-890c and finally washed with cold water.

J-BOX





Wool bleaching with H_2O_2

- The process of wool bleaching with hydrogen peroxide is near similar as cotton bleaching. But here P^H and temp. is somewhat lower.
- Recipe: To make 100 gallon liquor

1. H_2O_2 (35%) – 1.5 gallon
2. H_2O_2 (50%) – 1 gallon
3. Tetra sodium piro phosphate – 2 lbs
4. Etylene diamine tetra acetic acid – 1 lb
5. Water – The rest of
6. PH – 7.5 -9
7. Temp. – 500 – 550c
8. Time – 5 -6 hrs.

- The above chemical makes liquor in bleaching kier and immersed the woolen mtl. into liquor. After bleaching, all mtls are washed by 1-2% sodium bisulphate for 10 -20 minutes at room temp.

Silk bleaching with H_2O_2

- Near similar to wool bleaching but temp. is little high. In this case, Silk must be degumed before bleaching.
- **Bleaching recipe:**

1. H_2O_2 (35%) – 13 ml/L
2. H_2O_2 (50%) – 8 ml/L
3. Tetra sodium piro phosphate – 2 gm/L
4. Etylene diamine tetra acetic acid – 1 gm/L
5. P^H – 10
6. Temp. – 700c
7. Time – 2 - 4 hrs.

- For Tossar silk – mtls. are immersed in 0.6% H2O2 solution for 6hrs at 700c. After bleaching the mtls are washed by 1% sodium bisulphate solution for 15minutes at room temp. and then cold wash is done and dried.

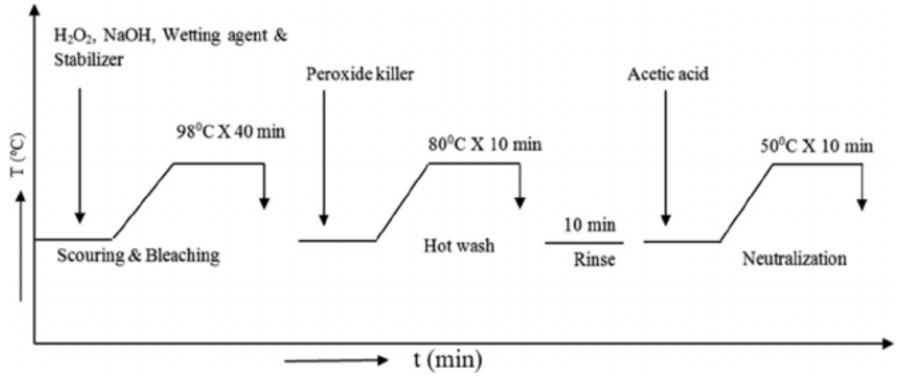
Jute bleaching with H_2O_2

- In this case, the wt. of jute decrease 20 -30% of its raw wt. Strength also lower. At first Sodium Chlorite bleaching and then hydrogen peroxide bleaching is done.

1. H_2O_2 (35%) – 0.7-1.4%
2. NaOH – 0.25%
3. Soda ash – 0.5%
4. Sodium silicate – 3% (78 Tw)
5. Magnesium sulphate – 0.1%
6. Temp. – 71-82^{0}c
7. Time – At first, circulation of liquor for 20-30 min then, 2-3 hrs.

- After bleaching at first hot wash then cold wash

COMBINED SCOURING BLEACHING CURVE

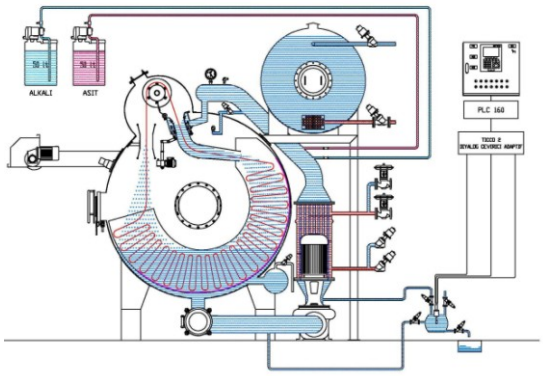


RECEIPE

- Fabric: X gm
- H_2O_2 : 4-5 gm/l
- Caustic soda: 4-5 g/l
- Sequestering agent: 1-2 g/l
- Peroxide stabelizers: 1-2 g/l
- Wetting agent:1-2 g/l
- M:L-1:8
- Time-1 hr
- Temp -80◦ C

WASHING RECEIPE

- Peroxide killer(PK)- 2-3 g/l
- Acetic acid-2-3 g/l
- Temp-50◦ C
- Time-30-60 min



MODERN WINCH DYEING MACHINE





